Empowering Women Through Radio: Evidence from Occupied Japan*

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Abstract

I study the impact of women's radio programs that the US-led occupying force aired nationwide in Occupied Japan (1945-1952) to dismantle the prewar patriarchal norms. From the perspective of the economics of identity, the radio messages can be viewed as attempts to alter gendered identity norms, and thus to shift women's political, economic and family outcomes. Using local variation in radio signal strength driven by soil conditions as an instrumental variable, I show that greater exposure to women's radio programs increased women's electoral turnout, and the vote share for female candidates, highlighting women's votes matter. I find no effects on women's labor market outcomes, but exposure to women's radio programs accelerated the postwar fertility transition. Overall, disseminating pro-gender-equality messages can have significant implications for both women's lives and society at large, potentially paving the way for rapid economic growth that would follow.

JEL codes: A13, D72, J13, J16

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1 Introduction

Over the last century, societies have made a significant stride in expanding women's socioeconomic opportunities. Even wars and armed conflicts, albeit terrible ordeals, have provoked changes in the economic landscape, which can further serve as catalysts for advancing women's economic as well as political participation. However, a growing body of economic research suggests that preexisting gendered norms and attitudes often persist, continuing to hold women back. As highlighted in the reviews by Giuliano (2020) and Lowes (2022), much of the literature has focused on the historical persistence of these norms. What remains less understood is which public policies have weakened these constraints and driven changes in women's political, economic, and family outcomes.

In this study, I examine the impact of pro-gender-equality messages disseminated through radio on women's outcomes in Occupied Japan (1945–1952). Beginning on October 1, 1945, the US-led occupying force (General Headquarters, Supreme Commander for the Allied Powers, or GHQ/SCAP) aired women's radio programs to challenge and dismantle deeply embedded patriarchal norms in prewar Japanese society (Japan Broadcasting Corporation 1947). My textual analysis of GHQ/SCAP Weekly Radio Reports reveals that these programs promoted women's empowerment across three key dimensions: political participation, economic inclusion, and bodily autonomy. Drawing on Akerlof and Kranton (2000)'s economics of identity, I predict that these broadcasts served as a mechanism for reshaping gender norms, and increased women's electoral and labor force participation while accelerating fertility decline.

To empirically investigate the impacts of radio messaging, I build a unique district-level dataset that covers both the prewar and postwar periods. The dataset includes data on election results, labor force participation, marriages, births, and the geographical reach of the radio, gathered from various historical resources such as the Japan Broadcasting Corporation's reports, maps, newspapers, official election results, newly digitized vital statistics, and population census. Exposure to women's radio programs, which is my main independent variable, comes from district-level variation in radio listenership. Listenership is proxied by the district-level radio subscription rate, a statistics that originates from the state-owned radio licensing system at the time. Across the nation, listenership varied from below 10 percent to above 70 percent with an average of 37 percent. This provides cross-sectional variation in women's exposure to messages regarding gender-egalitarian norms on air.

A key empirical challenge is that radio listenership may not have been randomly assigned. Areas with higher listenership could have differed systematically in ways that influence women's outcomes. To address this potential endogeneity, I employ an instrumental variable (IV) strategy that exploits variation in AM radio signal strength driven by local soil conductivity. Signal quality is generally determined by factors such as distance to a transmitter and transmission power, and I control for these directly. I then use the remaining variation in field strength, attributable to soil characteristics, as the instrument for radio listenership. This approach isolates plausibly exogenous variation in exposure to women's radio programs. The identifying assumption is that soil-driven differences in radio signal affect women's political participation, labor market activity, and fertility only through increased exposure to radio programs, and are not correlated with unobserved determinants of these outcomes. I support this assumption by showing that the IV is uncorrelated with a range of preexisting local characteristics, including measures of female empowerment, democratic penetration, attitudes toward the wartime military government, sectoral composition, and pre-U.S. occupation outcomes of interests.

My findings are three-fold. First, I find that greater exposure to women's programs significantly increases women's voter participation in the first postwar election where women were allowed to vote. Specifically, a one standard deviation increase in exposure to women's radio programs increases women's electoral turnout by 2.5 percentage points. Based on this estimate, the persuasion rate (DellaVigna and Gentzkow, 2010) is 18.1, which has a comparable size with, but larger than, the persuasion rates of mass media on voter participation in historical contexts (Gentzkow et al., 2011), and close to the rate achieved by effective Get-Out-the-Vote (GOTV) campaigns.

In areas with greater radio exposure, not only female voter participation, but also vote share for female candidates was greater: a one standard deviation increase in exposure also raises a female candidate's vote share by 2.3 percentage points. This positive effect contributes to women's increasing representation in the national legislature. By calculating the counterfactual election outcome with zero radio exposure in every district, I demonstrate that women's share among winners would have been only 5.3 percent, compared to the actual representation of 8.2 percent. This finding makes a case against the *double-vote hypothesis*, which posited that women, especially married women, would simply cast the same votes as their husbands and have no influence on electoral outcomes. My finding, by contrast, highlights that women's votes matter.

Second, I find that radio exposure *accelerates* the decline in the number of births. A one standard deviation increase in radio exposure contributes to decreasing the 1950 birth rate by 2.0 per 1,000 population. Around that time, Japan's birth rate began to decline as it entered a fertility transition,[1] following a brief postwar baby boom. My findings imply that radio messages contributed to speed up the fertility transition. Data show that the present effect is not likely due to increased abortions. I also provide supporting evidence that the decline in fertility is not simply a delay in births, but likely a decrease in the number of lifetime births per woman. Overall, my results imply that the empowerment of women contributed to the notably rapid fertility transition in postwar Japan.

Third, one aspect unaffected by the radio messages was women's participation in the labor market. This could be due to a tightened labor market following the influx of men returning from World War II. It could also be due to the persistent traditional attitudes of male-dominated employers and men's view on working women remained unchanged. This is plausible, considering that the radio campaigns were primarily directed at women, possibly overlooking the need to simultaneously change men's norms.

Further analyses suggest that the main findings are likely driven by empowering messages directed at women rather than by general radio exposure. Moreover, the impacts do not vary by women's education level or by whether districts are urban or rural, both factors that would typically reflect informational gaps. This indicates that information alone is not the primary channel, and instead the results point to the role of normative messages. To corroborate this interpretation, I provide correlational evidence showing that women's higher postwar radio listenership is associated with more egalitarian gender norms. Finally, the radio impacts are stronger in areas with lower male to female ratios, suggesting that a greater presence of women in the locality

[1] In 1947, the crude birth rate was 34.5 per 1,000 population and the total fertility rate was 4.54. By 1955, the crude birth rate had declined to 19.5 per 1,000 and the total fertility rate to 2.37, according to the National Institute of Population and Social Security Research.

amplifies the effects.

Contribution to Literature This study contributes to three strands of literature. First, this study provides evidence that mass media can help disseminate messages that women's voice counts in elections, which get them out to vote and influence the electoral outcomes. Generally, the literature on mass media and electoral turnout has shown that political information disseminated through mass media increases voter turnout (DellaVigna and Gentzkow 2010, Gentzkow et al. 2011, Strömberg 2004, Strömberg, 2015). This line of literature, however, mostly focuses on long-enfranchised voters. Meanwhile, my study shed light on the newly enfranchised women. They deserve careful attention. Previous studies suggest that "[o]verall turnout declined as a result of adding women to the eligible electorate." (Corder and Wolbrecht, 2016). While data on gender-specific turnout are rarely available, a few studies confirm that women turned out less than men did (Corder and Wolbrecht, 2016) when they were newly franchised across the globe. That is to say, women's suffrage, albeit a crucial milestone in advancing the position of women, may not automatically translate into equal exercise of political power between men and women. My findings highlight that disseminating new norms through mass media can be a way to move the needle: it mobilizes more women into the active electorate and influences electoral outcomes. Mass media help to incorporate women into the electorate.

Second, this study contributes to the economic literature on the impact of women's empowerment on fertility, particularly in settings where fertility rates remain high. Prior studies have shown that women's empowerment influences fertility through four economic channels: changing women's *opportunity cost* of time, enabling access to (Goldin and Katz 2002, Bailey 2006) and information (Kearney and Levine 2015) of contraceptives and reproductive *technologies*, enhancing women's relative *bargaining power* (Doepke and Kindermann 2019), and altering *public attitudes* and *norms* (see Silva and Tenreyro 2017 for review). The present study relates to the norm-transmission channel but differs in key ways from previous work. The radio programs I study directly targeted women and promoted birth control to improve women's health and empowerment. This contrasts with the dissemination of family planning through court rulings reported in newspapers studied by Beach and Hanlon (2022). While La Ferrara et al. (2012) shows that commercial entertainment television portrayed small families as role models and reduced fertility, the programs I study were publicly run by the occupation authority, making the findings more directly policy relevant. Whereas Jensen and Oster (2009) find larger effects of cable television in rural India with limited access to information, I do not find urban-rural differences, suggesting that information is not the likely channel. Finally, I find no evidence that radio exposure increased stillbirths, a proxy for abortions, which indicates limited unintended consequences. This is an additional aspect not previously examined in the literature.

Third, this study speaks to the economic literature that examines the impacts of radio broadcasting. Studies show that radio messages work by reducing information friction (Strömberg 2004, Svensson and Yanagizawa-Drott, 2009, Russo 2019), disseminating propaganda or populist persuasion (Yanagizawa-Drott 2014, DellaVigna et al. 2014, Adena et al. 2015, Wang 2021a), changing time use (Olken 2009), coordinating social movement (Gagliarducci et al. 2020, Wang 2021b), or facilitating norms and attitudinal changes (Cheung 2012, Blouin and Mukand 2019, Khalifa 2022). While the mechanism at work depends on the context, the norm channels appear to fit the present study. In terms of empirical strategies, this paper follows the spirit of Strömberg (2004) and Adena et al. (2015), which identify the impact of AM radio waves using ground conductivity as an instrumental variable. The empirical context is also unique, as the radio programs in this study were broadcast by a foreign occupier and implemented on a large scale. Additionally, this study provides a quantitative content analysis, which has been rarely undertaken in the existing literature due to limited data availability. .

Organization of the Paper The remainder of the paper is organized as follows. Section 2 provides a brief background of women's radio programs in Occupied Japan and highlights key features that are critical for my empirical analysis. Section 3 presents the hypothesis. Section 5 outlines the identification and estimation strategy, and Section 4 describes the data sources. Section 6 reports the empirical results. Section 7 considers the robustness of the findings, potential mechanisms, and the generalizability of the results. Section 8 concludes.

2 Historical Background

This section provides three elements of historical context essential to the study. Subsection 2.1 discusses how women's radio programs were among the Allies' first initiatives to raise women's social status. Subsection 2.2 describes the preexisting radio broadcasting infrastructure, which underpins the identification strategy. Subsection 2.3 analyzes the content of women's radio programs, both qualitatively and quantitatively, and examines women's political participation, labor market activity, and family formation as relevant behavioral outcomes.

2.1 The Allies' Efforts to Raise Women's Status in Occupied Japan

In prewar Japan, women lived under the influence of patriarchal norms. At school, they were taught the ideology of "Good Wife, Wise Mother (*Ryosai Kenbo*)." This concept refers to the ideal image of a woman at the time, rooted in the belief that a woman's primary duties are to manage the household, bear children, and raise them.[2] Meanwhile, women were excluded from politics – they were not only denied voting rights but were also prohibited from joining political organizations.[3] Upon marriage, women were considered "incapacitated." They needed their husband's permission to work, and their property was controlled by their husbands.[4] Their pursuit of birth control has also encountered resistance. The birth control movement began to gain momentum during the 1920s,[5] but the movement was suppressed when Japan entered the war in the 1930s. At that time, the complete fertility rate was 4.6 per woman. During the war, the Japanese government implemented pro-natal

[2]Studies show that textbooks for girls approved by the Ministry of Education adopted the ideology of "Good Wife, Wise Mother (*Ryosai Kenbo*)." (Koyama 1991, Jiang 2022)

[3]The Constitution of the Empire of Japan (*Meiji Constitution*), enacted in 1890, did not grant women voting rights. Moreover, the Association and Political Meetings Law of 1890 (*Shukai Oyobi Seisya Ho*) and the Article 5 of Public Safty and Police Law of 1900 (*Chian Keisatsu Ho*) barred women from joining a political organization or even just attending a political meeting. In 1922, the Article 5 of Public Safty and Police Law was amended in response to women's petition and women were allowed to attend a meeting. However, they continued to be barred from joining any political organization. The women's suffrage movement also gained momentum in the 1920s, yet it did not achieve success in the prewar period.

[4]The Articles 14-18 of the Civil Code of 1898 (*Meiji Minpo*).

[5]Baron Shizue Ishimoto (later Shizue Kato) started the birth control movement after she met an American birth control advocate, Margaret Sanger, in New York. However, this movement encountered resistance in the late 1930s.

policies to raise this number to over five. Marriages were encouraged by municipalities, married women without children were inspected, and parents with more than ten children were rewarded (Ogino 2008).[6]

After World War II, the Allied Occupation[7] prompted a seismic shift in women's status in Japan. The occupation's five major reform directives included raising the legal and social status of Japanese women.[8] GHQ/SCAP attributed prewar militancy to Japan's patriarchal system (Kobayashi 2004), and prioritized raising women's status as a key peacebuilding strategy. During the occupation period, Japanese women gained new legal rights, including rights to vote and run for office (in December 1945 for the national election, and in September 1946 for the local election), and greater access to higher education (1948). The new constitution, enacted in May 1947, has prohibited discrimination on the basis of sex.[9] Indeed, the U.S. occupying force "chose to make Japan a laboratory for one of the world's most radical experiments with women's rights" (Pharr 1987).

GHQ/SCAP viewed a change in women's mindset as crucial for raising women's status and, by extension, for the overall success of the Allied occupation. Therefore, their initial initiative was to launch radio programs targeting women. In fact, as early as October 1, 1945, just one month after the Allies started to occupy Japan, the government-sponsored radio station, the Japan Broadcasting Corporation, began to air programs targeting women. The women's programs aimed "to raise political, social, and cultural standards of ordinary women and breaking away from feudalism," and "in order to select qualified female leaders, [the women's radio programs intro-

[6] The Ministry of Welfare introduced the award (*Yuryo Tashi Katei Hyosho*) in 1940. In its first year, 10,336 parents received the award. The following year, an additional 2,134 parents were recognized.

[7] After World War II, Japan was occupied by the Allied Powers from September 2, 1945 to April 28, 1952. Although officially called the "Allied Occupation," it was mostly an undertaking of the United States, with contributions from Australia, India, New Zealand, and the United Kingdom, and therefore it was often called the "American Occupation." General Douglas MacArthur oversaw the occupation as the Supreme Commander for the Allied Powers (SCAP). The acronym SCAP was soon used to refer not only to the commander himself, but also to the offices of occupation set up under him to guide Japan to demilitarize and democratize the nation.

[8] The other four reforms were to abolish the secret police, to encourage the formation of labor unions, liberalize the education system, and democratize the economy. Source: Diplomatic Records A' 1.0.0.2-3-4 "Conference Abstracts and Memoranda between the Supreme Commander for the Allied Powers and his Staff and the Prime Minister and other ministers of Japan" <GAI-1, Reel No. A'-0055>.

[9] Uemura (2007)'s pioneering work uncovers the role of the Occupation regime in raising Japanese women's status and, in particular, analyzes the relationships between occupier and occupied.

duced] not only anti-militarists who remained silent during the war but also many unknown progressive, young women." (Japan Broadcasting Corporation Yearbook (1947); translated by the author).

2.2 Radio Reception and Use in Occupied Japan

At the onset of the Allied Occupation, there were 53 radio transmitters and 39 amplifiers across the nation, all of which were connected and operated by a single state-sponsored radio station, the Japan Broadcasting Corporation (JBC). The JBC has a primary channel, channel 1 (*daiichi hoso* in Japanese) which aired various programs throughout the day, and a secondary channel, channel 2 (*daini hoso*) which was used for only part of the day and mainly for rebroadcasting. Until 1952, there was no private radio broadcasting. In effect, a radio holder had only two choices: either to listen to the JBC's programs, or not to listen to any programs at all. Such a binary choice set turns out to be critical for my empirical analysis. I do not need to consider the listener's selection into different radio stations at the same time window.

Importantly for my analysis, the JBC kept records on the number of households subscribing to radio as well as the total number of households in all municipalities. This is because the JBC mandated all radio holders to register and pay subscription fees. The fee was not expensive and thus, I do not worry that it may have excluded low-income families from acquiring radio. In fact, the mandated annual fee in 1950 was 35 yen, which was 2.5 times the price of one serving of Soba noodles, Japanese staples.[10] I digitize the JBC record to calculate the radio subscription rate, which I later use as an independent variable in my empirical analysis.

During the Allied Occupation, the JBC operated under the close supervision of the GHQ/SCAP Civil Information and Education Section Radio Unit (later also called the Radio Branch and hereafter referred to as the Radio Unit). Radio broadcast content was censored in advance by the Radio Unit.[11] The Radio Unit also conducted

[10]See Table E.10 on annual radio subscription fee from 1925 to 1955. Data are drawn from Okabe (2018).

[11]The head of the production team for women's programs, Fuji Egami recalls "All women's programs, including introductory announcements, were to be submitted to the Civil Information and Education Section 10 days before the broadcast. Translators kept typing all the time. All dramas, stories, lectures, interviews, debates and even round tables were stenographed, rewritten into Japanese, then translated into English. It took more than two weeks to broadcast programs after

modern listener surveys (Mayo 1988, Luther and Boyd 1997, Smulyan 2002). In effect, the Radio Unit had a large say over what kind of content was on air and therefore played a key role in disseminating information to meet the GHQ's purposes.

The JBC started airing the flagship women's program "Women's Hour (*Fujin no jikan*)" on October 1, 1945, just about a month after the Allies started occupying Japan. The program was designed to draw as many women as possible. A time slot allotted to the women's program was the lunch break when women used to listen to a war-time women's program during World War II; Japanese women were appointed as the director of the production team, as well as moderators, in order to be friendly to female listeners; music was played occasionally for a pause so that listeners could maintain concentration (Japan Broadcasting Corporation, 1947, 1950)

As the JBC's Listener survey reveals, women's programs were indeed well received. In the 1947 survey, more than 70 percent of women with a radio subscription said that they currently listened to or used to listen to the women's program. Not only did they listen to the women's program, but more than 60 percent of them answered that they had gained new insights through the program.

As time went by and the JBC's production capacity increased, the JBC added more time slots for women's programs. By the end of 1950, the weekly airtime that the JBC allocated to women's programs had quadrupled from its onset in October 1945.[12] This underscores the fact that GHQ/SCAP maintained and strengthened efforts to raise women's status throughout the occupation. As the airtime expanded, the content covered by the women's programs grew too, as I show in detail in the next subsection.

Before diving into the radio content analysis, I note that the analysis primarily focuses on the occupation period (1945-1952) although the JBC continued to air women's programs until 1963. I restrict my attention mainly to the occupation period because, at the end of the occupation period, private radio broadcasting as well as TV broadcasting started, giving more choices to potential listeners. Competition among different mass media outlets may have fundamentally changed the nature of media

they were recorded. It was impossible to deliver timely information" (Egami 1955, translated in English by the author.)

[12] See Figure E.4 NHK Yearbook (Japan Broadcasting Corporation 1947, 1949) and GHQ/SCAP CIE Weekly Report (Radio Education Branch, 1946 - 1950).

content as well as complicated the listeners' decision process about which information they acquire and why. Although this transition in the broadcasting market opens up a new avenue of research, it is beyond the scope of my current analysis.

2.3 Content of Women's Radio Programs

What kind of messages did the women's programs try to disseminate? Answering this question is key to determine which women's outcome this study should consider. Therefore I turn to the Weekly Radio Reports (from January 1946 to December 1950), which document daily radio content. In the following, I first present some examples. Then, I classify the contents systematically into several topics and evaluate how the composition of these topics changed throughout the duration of the Allied Occupation.

The Weekly Radio Report[13] was reported every week in English, with one section dedicated to the featured programs of the week. The first example below sheds light on a significant change in norms concerning birth control:

> "*Should Birth Control Be Legalized?* (Women's Hour. January 31, 1946)
> Discussed the pros and cons on birth control, its effect on preservation of mother and child health; the necessity of spacing children in order to provide better care; reasons for and against legalization of abortions."

As mentioned in Section 2.1, the wartime government implemented pro-natal policies and discouraged birth control. The above example illustrates the significant shift in public discourse around birth control since then. It demonstrates the social acceptance of birth control for the sake of maternal and child health.[14] The next example challenges the patriarchal convention:

> "*The General Election and Man and Wife* (Women's Hour. March 12, 1946)
> While it may be praiseworthy for a wife to bow to her husband's will in

[13] Weekly Radio Report GHQ/SCAP CIE Box No. 5318 Folder 9.

[14] General Headquarters (GHQ) explicitly stated they would not interfere with Japan's population problems or promote birth control. Yet they did not prohibit discussions regarding birth control — especially from perspectives of women's liberation, maternal protection, and children's health (Yamamoto, 2017). This radio example further illustrates this point.

> many cases, the forthcoming general election demands that she make her own decision, entirely independent of others"

The program demands that women make their own choices of whom to vote. This goes against the conventional patriarchal view that a woman obeys her father, her husband, and her son. The final example then celebrates the improved status of women after World War II:

> "Working Woman's Hour, September 10, 1950.
> Equal rights for women in their working conditions was the subject of the principal feature on this program, in which Mrs.Hani advised working women to feel proud of their position among men where they are winning economic independence for the first time in Japanese history."

To understand the topic composition of the women's radio programs in a more systematic manner, I classify the daily contents of women's programs using Latent Dirichlet allocation, and show the year-by-year topic composition (Figure E.2).[15] I find that in 1946, women's programs primarily focused on politics and elections, which is consistent with histrians' account (Okahara, 2007). Over the years, the content of these programs became more diverse, covering topics such as women's organizations, the interests of young women and girls, child development, new labor and welfare laws, and information on food and health. Ultimately, the programs addressed three key aspects of women's empowerment - political, economic, and bodily autonomy.

3 An Economic Perspective on Women's Radio Programs

Through the lens of Akerlof and Kranton (2000)'s economics of identity, the radio messages can be viewed as a tool for reshaping and coordinating women's sense of self, namely gender identity norm, on a large scale. Women, previously seen as unfit for political participation, were now encouraged to be important political contributors; Discussions on gender equality in the workplace began to surface; Birth control, discouraged during wartime, was promoted as beneficial for maternal and child health. Through radio, listeners learned that traditional norms were being challenged and new

[15] For a detailed explanation, see Appendix E.2.

ones promoted. The medium created a shared experience in which women became aware that others were hearing the same messages. This collective awareness may have reduced the psychological costs of engaging in behaviors that had previously been seen as socially unacceptable. As a result, women exposed to these messages may have acted differently than they otherwise would have.

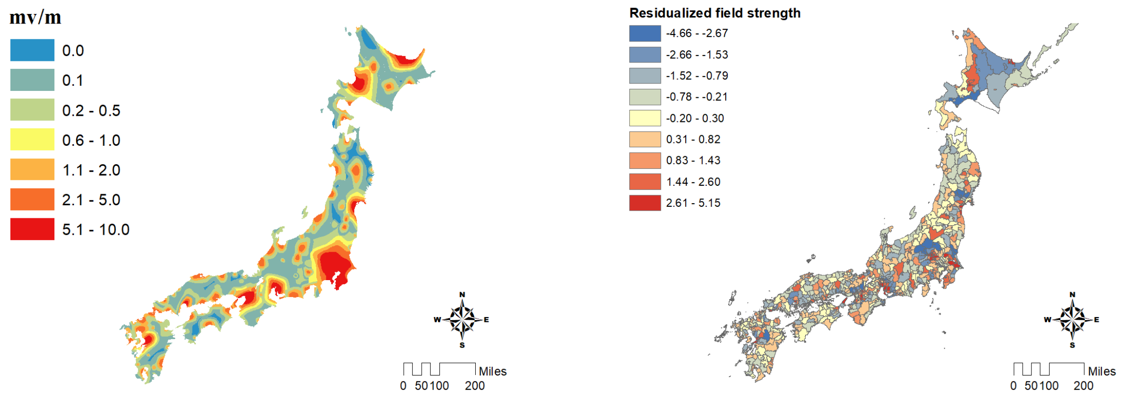
Given that women's radio programs covered three key aspects of women's empowerment, political, economic, and bodily autonomy, I hypothesize the following: First, women's electoral turnout increased with greater exposure to radio programs that urged women to vote in the 1946 general election. Second, women's labor market participation rose in response to increased exposure to radio content discussing women's careers and labor laws protecting women's workplace rights. Third, the annual birth rate decreased with more exposure to programs promoting the benefits of birth control and spacing for women's and children's health. However, it remains an empirical question whether these radio programs had the intended consequences. Internalized identity norms can be resistant to change, and radio campaigning might also backfire.

4 Data

I compile district-level data (*shi* and *gun*) on radio listenership, radio signal coverage, outcome variables, and other district characteristics from official election reports, population censuses, vital statistics, local newspapers, and publications of the Japan Broadcasting Corporation (JBC). The unit of observation is the 1950 district boundary, and data from other years are harmonized to this boundary, accounting for changes over time.[16] All variables except electoral turnout cover the entire nation.

The degree of exposure to women's radio programs is measured using the 1946 radio subscription rate, defined as the share of households with a radio subscription, based on data from the 1946 JBC yearbook. These figures come from Japan's radio receiver licensing system, which recorded district-level subscription rates. This measure is essential for scaling in the analysis, as subscription rates in 1946 ranged from about 5 percent to nearly 80 percent, with a national average of 37.3 percent.

[16] Small islands and districts with complex boundary changes are excluded from the sample.



(a) Radio 1 field strength (b) Residualized field strength

Figure 1: Field Strength and Residualized Field Strength

Notes: Panel (a) shows the AM wave field strength, which serves as the instrumental variable for radio exposure. I digitized the map that the JBC published in 1949. Panel (b) shows the spatial distribution of residualized field strength. The residualized field strength is obtained by regressing field strength on the distance to a nearby transmitter, transmitter fixed effects, and all the other district characteristics variables such as population density and industrial composition.

I construct an instrumental variable for radio exposure using AM wave field strength, digitized from a map published by the JBC in 1949 (Figure 1a). According to Japan Broadcasting Corporation (1947), the measurements were taken in 1948, making this the first postwar field strength map. I calculate average field strength by district and measure the distance to the nearest transmitter using latitude and longitude from the 1947 JBC yearbook[17] and shapefiles of administrative boundaries.[18]

I hand-collect turnout by sex for the April 10, 1946 House of Representatives election, which was the first after women's suffrage, from local editions of three national newspapers and multiple prefectural newspapers published between April 12 and April 18, 1946.[19] The dataset covers 662 districts in 26 prefectures, representing 54 percent of eligible voters.[20]

[17]Only two transmitters were added between 1946 and 1948 (from 111 to 113), so changes in transmitter location are unlikely to confound results.

[18]Shapefiles are from Maruyama Lab, Tsukuba University, Japan.

[19]Sources include the Yomiuri and its local editions in Iwate, Miyagi, Fukushima, Ibaraki, Gunma, Saitama, Chiba, and Nagano; the Asahi (April 11) and its local editions in Shizuoka, Shimane (April 18), Nagasaki, Oita, and Miyazaki; Yamagata Shinbun; Kitaguni Shibun; Yamanashi Nichi Nichi Shinbun (April 13); Chubu Nihon Shinbun; Ise Shinbun; Kyoto Shinbun; Bocho Shinbun (April 16); Tokushima Shinbun; Ehime Shinbun; and Saga Shinbun. Unless otherwise noted, the publication date is April 12, 1946. Shinbun means "newspaper" in Japanese.

[20]In 1946, all men and women aged 20 or above were eligible to vote and were automatically

Table A.2 reports summary statistics for the district-level radio subscription rate, residualized field strength (which provides as-good-as-random variation in radio signal due to soil conditions), and other radio-related characteristics for all districts (Column 1), districts in the election sample (Column 2), and districts outside the election sample (Column 3). All variables are statistically similar between in-sample and out-of-sample districts, which suggests no sample selection based on radio-related characteristics. In the later analyses, I focus on within-prefecture, across-district variation rather than variation across prefectures.

Other variables, including electoral turnout, election results, labor market participation, marriage, and fertility outcomes, are drawn from population censuses, vital statistics, and official election reports. In Table 1 Panel B Columns 5–7, I report prewar mean outcomes: the female employment share is 0.38 (excluding family employment), the marriage rate is 9 per 1,000 population, and the birth rate is 33 per 1,000 population. Table A.1 lists all data sources.

5 Empirical Strategy

To examine the impacts of women's exposure to women's radio programs, I use a linear causal model relating exposure to the outcomes of interest at the district level. For a district j in period t,

$$y_{j,t} = \beta_0 + \beta_1 r_j + \gamma x'_{j,t} + u_{j,t} \tag{1}$$

where $y_{j,t}$ is an outcome of interest, r_j is the degree of exposure to women's radio programs, and $x_{j,t}$ is a vector of district characteristics, such as the distance from the nearest radio station, radio station fixed effects, prefecture fixed effects, population size, and to the extent of damage (casualties) caused during World War II.

As explained in Section 4, I measure the key explanatory variable, women's exposure to women's radio programs (r_j, expressed in standard deviation units), using the district-level radio subscription rate, which is defined as the share of households with a radio subscription. Potential measurement concerns are addressed in Section 7.2. The main outcome variables are female voter turnout and the vote share of female

registered.

candidates in the 1946 election, the birth rate in 1947 and 1950, and the female employment rate, marriage rate, and divorce rate in 1950. The coefficient on the degree of radio exposure, β_1, is the parameter of interest and captures the effect of exposure to women's radio programs on each outcome.

The key empirical challenge to identify the causal effect β_1 is that radio exposure might be endogenous to women's outcomes through unobserved district characteristics. It would be possible that radio subscriptions are correlated with subscribers' unobserved characteristics, such as attitude toward female empowerment and/or the Allied Occupation, openness to new ideas, or local culture. For example, women with a greater interest in politics may subscribe to radio to obtain information on politics. Such a positive correlation between subscription and unobserved characteristics bias the causal effect β_1.

To address the potential endogeneity issue, I instrument the exposure by quasi-random variation in radio signal quality during daytime hours, which is as good as random to potential subscribers but increases their likelihood of radio subscription. The metric of the radio reception I use is the ground wave field strength (hereafter field strength), which depends on the horizontal distance from a nearby transmitter, output power of the transmitter, wavelength, and ground conductivity between the transmitter and receiver. The ground conductivity measures how fast the AM radio wave can propagate through a given soil type and depends on the moisture and salt contents of the soil. The key idea is as follows. On the one hand, the distance to a nearby transmitter, output power, or wavelength may be based on strategic considerations,[21] and the ground conductivity is as good as random to potential subscribers. Therefore, after controlling for distance and transmitter fixed effects, the local variation in the field strength can serve as an instrumental variable for the radio subscription rate. My identification strategy is similar to Strömberg (2004) and Adena et al. (2015) which also exploits features of AM radio wave propagation. Thus,

[21] For example, radio transmitters may have been strategically placed in the area with higher political aspirations, higher aspiration for freedom of marriage, higher demand for birth control, higher potential supply of female labor force, and so forth. Although such a concern may be unwarranted given the historical background of radio, as described in Section 2, these unobserved characteristics of women may indirectly relate to transmitter locations through urbanness, and therefore it is still important to control for the distance.

I run the first stage equation as follows:

$$r_j = \pi_0 + \pi_1 z_j + f\big(d_{j,c(j)}\big) + \delta_{c(j)} + \phi_{p(j)} + x_j'\eta + \nu_j \tag{2}$$

where z_j denotes field strength, $d_{j,c(j)}$ is the distance from district j to its nearest transmitter, $\delta_{c(j)}$ captures fixed effects of that transmitter, $\phi_{p(j)}$ is prefecture fixed effects,[22] and $\delta_{c(j)}$ includes the log of the number of households in district j, as well as an indicator equal to one if the district was severely damaged by air raids or the atomic bomb during World War II. I control for distance nonlinearly by including dummy variables for each distance decile.

Figure 1b shows the identifying variation, namely the residualized field strength after controlling for distance, transmitter fixed effects, and all other control variables. Areas with high residualized field strength, shown in dark blue, and those with low values, shown in dark red, are located next to each other. This visual pattern suggests no systematic spatial structure, supporting the assumption that the residualized field strength is as good as randomly assigned. As shown in Figure 2, higher residualized field strength is associated with higher radio subscription rates, demonstrating the relevance of the instrument. More detailed validation is provided in the next section.

With the instrumental variable in hand, I estimate the causal impact of radio exposure on each outcome using two-stage least squares (TSLS) with robust standard errors. The second-stage regression is specified as

$$y_{j,t} = \beta_0 + \beta_1 \hat{r}j + g\big(d_{j,c(j)}\big) + \delta_{c(j)} + \phi_{p(j)} + x'j\kappa + uj, t \tag{3}$$

where $\hat{r}j$ is the predicted radio exposure from the first stage regression (2). The term $g\big(d_{j,c(j)}\big)$ controls for distance to the nearest transmitter by including dummy variables for each distance decile. $\delta_{c(j)}$ are nearest-transmitter fixed effects, $\phi_{p(j)}$ are prefecture fixed effects, and x includes population size and the extent of wartime damage (measured by casualties). Standard errors are clustered at the transmitter level. The coefficient β_1 is the parameter of interest and measures the effect of a one standard deviation change in radio exposure, as predicted by residualized field strength, on the outcomes.

[22] In Japan, a prefecture is the primary administrative division, governed by an elected governor, legislature, and supporting bureaucracy. Districts are smaller geographical units within prefectures.

Table 1: Balance Test

Panel A

	Political variables pre WWII				Political variables during WWII	
	(1) Turnout (1928)	(2) WSL district (1928)	(3) WSL vote share (1928)	(4) Turnout (1937)	(5) Turnout (1942)	(6) IRAA vote share (1942)
Residuals	0.0138	-0.0212	-0.0537	0.0173	0.00537	0.00879
	(0.0138)	(0.0373)	(0.0782)	(0.0210)	(0.00890)	(0.0162)
Mean Dep. Var.	0.82	0.09	0.10	0.78	0.85	0.68
Observations	662	662	60	638	662	662

Panel B

	Demographic and Economic Variables pre/during WWII				Pre US Occupation Outcomes		
	(1) Male to female ratio at birth (1935)	(2) Male to female ratio (1940)	(3) Gender industrial segregation (1940)	(4) Primary sector labor share (1940)	(5) Female Employment Rate (1940)	(6) Marriage rate (1935)	(7) Birth rate (1935)
Residuals	-0.580	-0.654	-0.00144	-0.0152	-0.00328	0.159	0.233
	(0.479)	(0.412)	(0.00486)	(0.0621)	(0.0166)	(0.246)	(0.987)
Mean Dep. Var.	105.25	97.95	0.08	0.55	0.38	8.56	32.71
Observations	662	658	658	658	658	662	662

Note: WSL – Women's Suffrage League; IRAA – Imperial Rule Assistance Association. This table presents regression results examining the relationship between pre-war district characteristics and the residual radio signal. The residual signal is constructed by regressing radio field strength on decile-binned distance to the nearest transmitter, radio station fixed effects, prefecture fixed effects, the logarithm of the number of households, and an indicator for severe WWII damage from air raids or the atomic bomb. This residual captures variation in radio signal strength attributable to soil conditions between the transmitter and the receiving district. The unit of observation is a district (*shi* or *gun*), using 1950 boundaries. In Panel A, Column 3 restricts the sample to districts with at least one WSL-endorsed candidate. Column 4 restricts the sample to districts where voting occurred, excluding districts where the number of candidates equaled the number of seats. All regressions are weighted by district population, and standard errors are clustered at the radio station level. See Appendix B for variable definitions and data sources.

6 Findings

Instrument Validity The IV strategy in Section 5 requires that the instrument (the residual variation in the field strength) is uncorrelated with unobserved determinants of the outcome (exogeneity) and influences the outcome only through increased exposure to radio programs (excludability). A key concern is that radio transmitters may have been placed in districts whose characteristics also predict the outcomes of interest. For example, these districts may have been more politically active, held different gender attitudes, or had varying views toward the military government and, by extension, the U.S. occupation. However, this type of strategic placement is less likely, as transmitters were not installed by the U.S. occupying authority, as explained in Section 2.2.

To further assess validity, I test whether pre-U.S. occupation district characteristics are correlated with the residualized radio signal. I focus on four potential confounding pathways: (i) gender attitudes and women's empowerment, (ii) democratic practices, (iii) support for the wartime militaristic government, and (iv) economic development. Below I summarize the approach; details on data sources and variable construction are in Appendix B.

First, districts with stronger signals might differ in gender attitudes and women's empowerment. To proxy this, I construct several indicators: (a) whether a male candidate publicly supported women's suffrage in the 1929 election and his vote share; (b) the sex ratio in reported births from the 1935 vital statistics, reflecting son preference; and (c) an industrial sex segregation index measuring the extent of gender overlap in non-home employment. Second, districts with stronger signals may have had more established democratic practices, potentially shaping turnout through habit formation. I proxy this with district-level turnout in the 1928 general election (the first after male universal suffrage) and the 1937 election (the last before World War II). Third, districts with stronger signals may have varied in their support for the wartime militaristic government, shaping attitudes toward the U.S. occupation. I proxy this with turnout and the vote share of the Imperial Rule Assistance Association in the 1942 wartime election, when existing parties were replaced to mobilize war support. Fourth, districts with stronger signals may have been at different stages of economic development, particularly in their reliance on primary industries. I proxy this with the labor share in agriculture, fishing, and mining from the 1940 Population

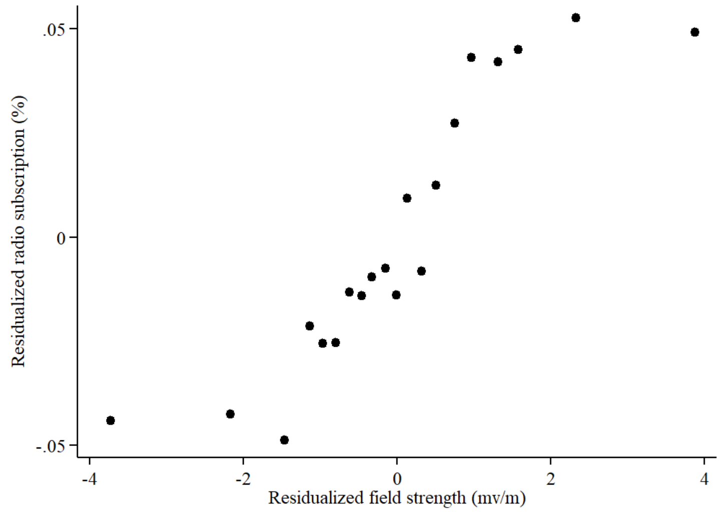


Figure 2: Visualization of the First Stage

Notes: This figure shows a binscatter plot of the residualized radio subscription rate against the field strength with a minimum set of control variables, i.e, the distance to a nearby transmitter and transmitter fixed effects. Each bin represents about 40 districts. This figure is a graphical depiction of the first stage: as we expect, greater residualized field strength does increase radio subscriptions. The S-shaped curve implies that the extremely low [high] field strength does not affect radio subscriptions because people there cannot [can] listen to radio anyway.

Census.

Table 1 shows that none of these variables are significantly correlated with residualized field strength in pairwise comparisons. In a joint regression, the covariates explain only 3 percent of the variation ($R^2 = 0.03$), and no coefficients are significant (Column 3 of Table B.3). Repeating the exercise on the electoral subsample yields the same conclusion: no significant correlations with residualized signal strength (Column 1). Taken together, these results provide support for the validity of the instrument.

First Stage Figure 2 shows a binscatter plot of the residualized radio subscription rate against the residualized field strength, showing a graphical representation of the first stage of my IV model. Column 1 of Table 2 reports the first stage estimate with the minimal set of covariates: after controlling for the transmitter fixed effect and distance to a nearby transmitter, a one standard deviation of field strength increases radio subscription by 0.57 standard deviation unit.

Impacts on Voter Participation I first present the results of voter participation in 1946 to illuminate the immediate impact of radio exposure. I find that exposure to women's radio programs yields positive impacts on women's political participation. Table 2 Column 4 shows a two-stage least-square estimate of the coefficient on the radio exposure in regression equation (1). A one standard deviation increase in radio subscriptions increases women's electoral turnout by 2.5 percentage points.

Meanwhile, Column 7 presents the TSLS estimate regressing men's turnout on the radio subscription. Similar to the OLS estimate, the TSLS estimate is not statistically distinguishable from zero at the five percent level. The positive gap between women's turnout (Column 5) and men's turnout (Column 7) suggests that the radio subscription has impacts only on women through the provision of women's radio programs.

The estimated impact is both statistically and substantively significant: it explains 31 percent of the standard deviation in women's turnout. The implied persuasion rate (DellaVigna and Kaplan, 2007), which is the share of eligible voters who changed their behavior due to radio exposure among those at risk of changing, is 18.1%.[23] For comparison, prior studies estimate persuasion rates of mass media on turnout in historical contexts between 4.4% for television (Gentzkow, 2006) and 12.8% for local newspapers (Gentzkow et al., 2011). The effect I find is larger than these historical benchmarks and closer in magnitude to the impacts typically achieved by effective Get-Out-the-Vote (GOTV) campaigns: 15.6% for door-to-door canvassing (Gerber and Green, 2000) and 20.4% for phone calls by youth volunteers (Green and Gerber, 2001) (see DellaVigna and Gentzkow 2010 for review). One likely reason for the relatively large effect in my context is that newly enfranchised women lacked prior voting histories, making their turnout decisions especially malleable.

Impacts on Vote Share Table 3 reports the impacts of increased radio exposure on vote shares of female candidates. As Column 3 shows, in areas with greater radio exposure by a one standard deviation received more vote by 2.3 percentage points. The estimate implies a persuasion rate of 4.6%.[24] This falls within the

[23]The share of eligible voters who turned out because of radio exposure is 0.07, and the share of voters who were at risk of changing their behavior is 0.41.

[24]Following DellaVigna and Kaplan (2007), I compute the persuasion rate for vote share as the proportion of votes cast due to radio exposure, relative to the share of votes at risk of being

range documented in previous studies, which vary from 2.0% for the unsurprising Democratic endorsement on support for Gore to 11.6% for the availability of Fox News on Republican vote share (see DellaVigna and Gentzkow 2010 for review).

One can calculate the counterfactual election outcome with zero radio exposure in every district. Women's share among winners would have been only 5.3 percent, compared to the actual representation of 8.2 percent. This finding makes a case against the *double-vote hypothesis*, which posited that women, especially married women, would simply cast the same votes as their husbands and have no influence on electoral outcomes. My finding, by contrast, highlights that women's votes matter.

Employment and Family Formation Turning to the effects on women's labor force participation, Upper Panel Column 2 of Table 4 shows that the estimated impact is statistically indistinguishable from zero.

Finally, I turn to the impact of women's programs on family formation. As Table 4 shows, both impacts on the marriage rate (Upper Panel Column 5) and divorce rate (Column 8) are near zero and statistically indistinguishable from zero.

Radio exposure does have impacts on birth rate, which gradually emerges. In 1947, the estimated impact of a one unit standard deviation radio exposure is -0.7 births per 1,000 population, albiet statistically significant. In 1950, five years after the radio programming started, a one unit standard deviation radio exposure affects the crude birth rate by 2.0 (Table 4 Lower Panel Column 5), which is statistically significant at one percent.

Additionally, I show that the decline in live births is not due to the change in stillbirths (fetal deaths). One may worry that radio exposure may change stillbirths while the total births remain constant. For example, radio exposure may raise women's health awareness, which may prevent fetal deaths. Alternatively, radio exposure may inform women about the Eugenic Protection Law of 1948. Since the law legalized abortions, informed women may be more likely to access abortions. To best address such a concern, I examine the impact of radio exposure on stillbirths, which include both induced abortions and stillbirths due to natural causes after 12 weeks

persuaded, adjusted for overall voter turnout.

of pregnancy.[25] The lower panel of Table 4 Column 7 shows that the impact on the number of stillbirths per 1,000 births is 0.1 and not statistically significant.[26] Thus, the decline in live births is not likely due to the significant change in stillbirths.

Summary Women's radio programs contemporaneously amplified women's voices in the political sphere in Japan: the women's programs effectively lead more women to vote, which in turn resulted in closing the gender representation gap. My findings are also consistent with what the GHQ/SCAP Radio Unit wrote in a weekly radio report: the women's programs "undoubtedly contributed in a large measure to the fact that 65 percent of the eligible women voters went to the polls." [27]

Radio exposure has also played a role in accelerating fertility decline. The results above suggest that the decline is due neither to an increase in women's labor force participation, to a decrease in marriages, or an increase in the utilization of induced abortions. Therefore, the fertility decline can be attributed to the remaining mechanism, that is, increased family planning. This decrease can be seen as beneficial for women as it reduces health risks and child-rearing burdens. Furthermore, it holds significant implications for society at large.

The null result on female employment could be due to a tightened labor market following the influx of men returning from World War II. It could also be due to the persistent traditional attitudes of male-dominated employers and men's view on working women remained unchanged. This is plausible, considering that the radio campaigns were primarily directed at women, possibly overlooking the need to simultaneously change men's norms.

[25] No data available on abortions and stillbirths due to natural causes separately.

[26] The number of births is the sum of the number of live births and stillbirths. Data on stillbirths are not available for the year 1947.

[27] Weekly radio report, SCAP Civil Information and Education Section.

Table 2: Impacts of Women's Radio: Voter Participation

	Radio subscription in std.dev. unit	Women's turnout			Men's turnout		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	FS	RF	OLS	TSLS	RF	OLS	TSLS
Field strength in std.dev. unit	0.571***	0.0139*			0.00514		
	(0.105)	(0.00587)			(0.00520)		
Radio subscription in std.dev. unit			0.0257***	0.0244**		-0.000272	0.00934
			(0.00555)	(0.00839)		(0.00451)	(0.00804)
Observations	337	337	337	337	327	327	327
First stage F-stat				29.33			28.16
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome		0.67	0.67	0.67	0.81	0.81	0.81

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: FS = first stage; RF = reduced form; OLS = ordinary least squares; TSLS = two-stage least squares. This table reports first stage, reduced form, OLS, and TSLS estimates of the effect of radio exposure on women's turnout (Columns 2–4) and men's turnout (Columns 5–7) in the 1946 general election. Column 1 shows the first-stage effects on radio subscription rates. The sample for men's turnout is smaller because in one prefecture (Miyazaki), the local newspaper reported turnout rates only for women, not men. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the nearest radio station level.

Table 3: Impacts of Women's Radio: Female Candidates' Vote Share

	Female Candidate's Vote Share			
	(1) RF	(2) OLS	(3) TSLS	(4) TSLS
Field strength in std.dev. unit	0.0143*			
	(0.00684)			
Radio subscription in std.dev. unit		0.00488	0.0230*	
		(0.00323)	(0.00986)	
Women's turnout %				0.00711
				(0.00455)
Observations	618	618	618	618
First stage F-stat			19.10	9.75
Candidate FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes
Mean outcome	0.08	0.08	0.08	0.08

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: RF = reduced form; OLS = ordinary least squares; TSLS = two-stage least squares. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the candidate level.

Table 4: Impacts of Women's Radio: Employment, Marriage, Divorce, and Birth Rate

	Women's employment rate excld. family emp			No. marriages per 1000 population in 1950			No. divorces per 1000 population in 1950		
	(1) OLS	(2) TSLS	(3) RF	(4) OLS	(5) TSLS	(6) RF	(7) OLS	(8) TSLS	(9) RF
Radio subscription in std.dev. unit	-0.00168	-0.00630		-0.0899	0.0289		-0.0106	0.0158	
	(0.00348)	(0.0113)		(0.0944)	(0.178)		(0.0273)	(0.0348)	
Field strength in std.dev. unit			-0.00342			0.0156			0.00852
			(0.00694)			(0.108)			(0.0216)
Observations	662	662	662	660	660	660	659	659	659
First stage F-stat		36.41			34.22			34.08	
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.28	0.28	0.28	8.42	8.42	8.42	0.97	0.97	0.97

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

	No. live births per 1000 population in 1947			No. live births per 1000 population in 1950			No. still births per 1000 population in 1950
	(1) OLS	(2) TSLS	(3) RF	(4) OLS	(5) TSLS	(6) RF	(7) TSLS
Radio subscription in std.dev. unit	-0.355	-0.698		-1.249***	-2.034***		-0.0788
	(0.209)	(0.450)		(0.341)	(0.514)		(0.115)
Field strength in std.dev. unit			-0.379			-1.103**	
			(0.257)			(0.333)	
Observations	662	662	662	662	662	662	661
First stage F-stat		36.41			36.41		34.15
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	34.54	34.54	34.54	28.14	28.14	28.14	2.64

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: OLS = Ordinary Least Squares; TSLS = Two-Stage Least Squares; RF = Reduced Form. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the nearest radio station level.

7 Discussion

Overall, my findings indicate that greater exposure to women's radio programs influenced women's political outcomes and fertility during the Allied Occupation. In this section, I discuss potential mechanisms, examine possible longer-run effects beyond the Occupation period, and consider the generalizability of the results.

7.1 Robustness

To assess the robustness of my main findings, I re-estimate the main specification controlling for prewar outcomes, as well as for prewar outcomes together with additional district-level demographic characteristics. These characteristics capture prewar economic development and gendered outcomes, including the sex ratio at birth, industrial sex segregation, the overall sex ratio, and the primary-sector labor share. Results are reported in Tables C.4 and C.5, and remain robust overall.

I also re-estimate the main specification using spatial standard errors following Conley (1999). This approach allows for spatial correlation that decays with distance, using a weighting function defined by a kernel in each dimension of the Cartesian plane (north–south and east–west). The kernel equals one at the origin, decreases linearly to zero at a cutoff, and remains zero beyond that point. I measure economic distance by physical distance and set the cutoff at 30 km, approximately the average distance between neighboring districts. Results, reported in Table C.6, show that the main findings remain robust when accounting for spatial correlation.

7.2 Mechanisms

In Section 3, I have proposed that women's radio programs that broadcast empowerment-oriented messages could influence women's outcomes by reshaping gender identity norms on a large scale. In this subsection, I present correlational evidence linking radio exposure to gender norms, and consider two alternative interpretations by examining heterogeneous effects across districts with different male-to-female ratios, which varied significantly due to men's wartime participation and casualties.

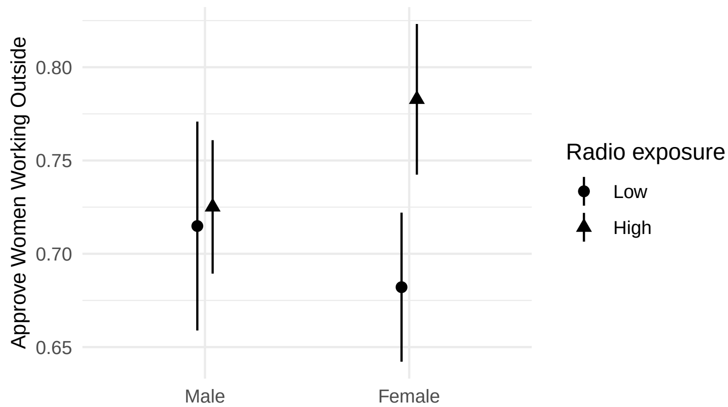


Figure 3: Radio Exposure and Gender Norms by Sex

Notes: The figure shows the predicted probability of approving women working outside the home by gender (x-axis) and level of radio exposure (distinguished by marker shape). Each point corresponds to the model-predicted approval rate for the given subgroup, and the vertical bars show 95% confidence intervals around those predictions. The regression result is presented in Appendix E.6.

Radio Exposure and Gender Norms In Appendix E.6, I present correlational evidence that frequent radio exposure is positively associated with gender-egalitarian views (approval of women working outside the home) among women but not among men. Figure 3 provides a visual summary of this pattern. The survey was conducted in 1953 by the Institute of Statistical Mathematics in Japan, just a year after the U.S. occupation ended. It is the earliest survey I could find that asked about both radio listenership and gender attitudes, with micro-level data available.[28] Although the evidence is correlational, the results indicate that women who reported frequent radio exposure were more likely to endorse egalitarian views, whereas men's views were unaffected. This pattern is consistent with the original hypothesis that radio exposure contributed to shifts in women's gender norms.

Women's Program versus General Program Since radio exposure is measured using subscription rates, one alternative interpretation is that the estimated effects may capture exposure to general radio programming, such as daily news, rather than content specifically aimed at women.

However, this possibility appears unlikely. As shown in Table 2 Column 7, radio

[28]Because the survey lacks information on respondents' place of residence, I cannot use location-based variation in field strength as an instrument for radio listenership.

exposure has no significant impact on men's turnout. This serves as a useful test for the influence of general programming. Since women's programs were designed for female audiences and men listened to them far less frequently, if at all (see Figure E.3), the absence of an effect among men suggests that general radio content alone did not substantially mobilize voters.

To extend this logic and conclude that women, too, were unaffected by general radio programming, we must assume that men and women were equally susceptible, or equally unresponsive, to general information transmitted via radio. This assumption could be challenged if newly enfranchised voters were systematically more responsive to radio, since in 1946 all women were voting for the first time, whereas men over age 25 had been eligible to vote since the prewar period.

Yet, my data suggest this concern is not warranted in the context I study. In Appendix D, I examine whether radio exposure increased men's turnout in the 1928 election, which was the first general election held after all adult men gained voting rights. I find no effect, indicating that newly enfranchised men in 1928 were no more responsive to radio exposure than experienced voters. This finding reduces the concern that newly enfranchised voters are particularly malleable to radio influence, at least in the Japanese electoral context of the time.

Taken together, the evidence supports the interpretation that the primary impact of radio on women's turnout in 1946 was driven by the content of women-specific programming. This is perhaps unsurprising, given that women's programs, unlike general broadcasts, covered topics highly relevant to women, such as profiles of female candidates, labor laws affecting women, the right to choose marriage partners, and the health benefits of birth spacing.

Normative Message versus Informational Content While the women's radio programs intended to dismantle the existing patriarchal norms, there is also a possibility that radio may have provided new information to women. The informational channel would be particularly probable if the radio campaign affected more illiterate and less-educated women, who were more informationally constrained. To assess whether the radio impacts differ based on the district's level of educated women, I estimate the following regression:

$$y_{j,1946} = \beta_0 + \beta_1 r_{j,1946} + \beta_2 r_{j,1946} \times z_j + \beta_3 z_j + \gamma x'_{j,1946} + u_{j,1946} \tag{4}$$

where z_j is an indicator equal to one if the share of highly educated women (10 or more years of education) is above the median. I instrument radio exposure with field strength. As shown in Figure C.1, the average residual field strength does not differ between districts with above-median and below-median shares of highly educated women.

The third rows of Table C.7, Column 2, and Table C.8, Column 2, present the heterogeneous effects of women's radio programming on turnout and fertility, respectively, by the share of highly educated women in a district. The effect of radio exposure on turnout does not vary with women's educational attainment. For fertility, a higher share of highly educated women slightly amplifies the radio effect, but not vice versa. These results do not support the hypothesis that radio primarily served to inform illiterate or less-educated women.

Similarly, I test whether the impacts differ between urban and rural districts by setting z_j in equation (4) to an indicator equal to one if district j is a city. This heterogeneity analysis examines whether radio exposure might have had a larger effect in rural areas ($z_j = 0$), where access to information may have been more limited. However, I find no statistically significant heterogeneous effects (see the 5th row of Table C.7, Column 2 for turnout, and Table C.8, Column 2 for fertility). Again, this result provides little support for the view that the informational content of women's radio programming is the primary driver of my main findings, and instead favors the original hypothesis that normative messages aimed at dismantling pre-war patriarchal norms plays a key role.

Male to Female Ratio Finally, I explore the possibility of heterogeneous impact based on the male to female ratio. Due to military casualties, the male to female ratio varies substantially with the mean of 0.92 and the standard deviation of 0.06. Asai and Kambayashi (2023) document that such across-district variation comes from *home regiments*: each regiment consisted of men from the same prefecture and it was as if home away from home. Moreover by chance, some regiments were heavily damaged. This is because the allies employed the leapfrogging strategy, i.e., bypassing heavily fortified islands instead of capturing every island in sequence. Therefore, variation in the male-to-female ratio in the aftermath of WWII is as-good-as-random, and provides the opportunity to test the heterogeneous effect of radio exposure.

Different theories predict heterogeneous impacts in opposite directions. On the one hand, marriage bargaining theory suggests that female scarcity should yield a positive coefficient β_2. The reasoning is that when women are scarce, their bargaining power increases, which could amplify the effect of radio exposure. Whether this mechanism operates in the present setting is ultimately an empirical question. On the other hand, prior studies such as Grosjean and Khattar (2019) show that male-biased sex ratios tend to sustain conservative gender norms. If this mechanism dominates, male abundance would suppress the effects of radio in areas with higher male-to-female ratios.

The seventh rows of Table C.7, Column 4, and Table C.8, Column 4, report the heterogeneous effects of women's radio programming on turnout and fertility, respectively, by the male-to-female ratio. A one standard deviation increase in the male-to-female ratio reduces the impact of radio exposure by 2.7 percentage points. In other words, radio exposure has a smaller effect on women's turnout when men are more abundant and women are relatively scarce. For fertility, the point estimate is in the same direction, although not statistically significant. Overall, these findings are more consistent with the mechanism that male abundance sustains conservative gender norms than with the marriage bargaining theory.

7.3 Possibilities of Long-Run Consequences

Beyond the immediate impacts, it is relevant to ask how the effects evolved in the medium and long run. One question is whether the mid-run effects of the radio programs were sustained, faded, or amplified. Using qualitative evidence and aggregate quantitative data, I suggest that the impacts were complex. Appendix E.4 shows that women's turnout increased in the 1949 and 1952 House of Representatives elections, with low-exposure areas catching up to high-exposure areas, likely due to spillovers and social sanctions. At the same time, qualitative accounts suggest resistance to women's growing political presence. MacArthur reportedly "warned them [female parliamentarians] from forming a female bloc" (Ichikawa 1977; Ogawa 1997), and an electoral reform in 1947 was expected to disadvantage women's representation.[29]

Long-run and intergenerational effects also represent an avenue for future research beyond the scope of this study. Women who listened to the radio programs may have raised their daughters with different expectations, and sons could likewise have served as an important channel. In the United States, Fernández et al. (2004) show that the wives of men whose mothers worked during World War II were significantly more likely to work, suggesting that norms can be transmitted through family dynamics and shape the next generation of women. A similar mechanism may have operated in my context. Examining such effects, however, would require contemporary individual-level data with detailed childhood geographic information, which are currently unavailable. Access to such data could shed light on radio's influence on marriage timing, partner matching, fertility, birth spacing, and regional migration patterns. Japan's rapid post-Occupation economic growth also presents an interesting case for studying the relationship between women's empowerment and economic growth, where the evidence remains mixed and further research is warranted (Duflo 2012; Anderson 2022).

7.4 Generalizability

Finally, my findings offer broader implications in two directions, though with important caveats. First, they speak to the potential of mass media to empower women and other underserved populations. While radio is a legacy medium, it remains widely used and influential in many parts of the world today. Thus, the core insights of this study would remain relevant.

That said, the context of Occupied Japan is unique as is always the case in an empirical study. Female empowerment is multidimensional (Anderson, 2022), and the radio programming I study delivered bundled messages spanning education, work, and civic life. This message bundling may have contributed to the observed impacts. Furthermore, postwar Japan was undergoing a profound societal rethinking, with

[29]For example, during the March 28, 1947 committee meeting, Kato Chuzo of the Social Democratic Party of Japan warned that the reform would reduce female representatives, and Ono Banboku of the Japan Liberal Party reportedly told women parliamentarians it might be their last chance to serve (The House of Representatives 1947a). Hara Hyonosuke raised the same concern on March 30, 1947 (The House of Representatives 1947b).

citizens reevaluating prewar norms. Such a moment of ideological openness may have made listeners particularly receptive to new messages, enhancing the efficacy of the broadcasts.

It is also important to note that the women's radio programs I study targeted women exclusively. To achieve greater and more sustained impacts, it may be necessary to engage men and the broader environment surrounding women. If pro-gender-equality messages had been directed at both women and men, they might have reinforced each other, although they could also have provoked backlash. One-sided messaging may also have contributed to diverging gender attitudes between men and women, potentially generating conflict. Women's radio programs were widespread internationally,[30] but their design and audiences varied. An instructive comparison comes from Sweden, a country now known for its gender egalitarianism, where in the 1960s campaigns promoted married women's employment not only through radio but also through evening television programs that families could watch together. These broadcasts explicitly linked women's work outside the home to changes in family organization (Lundqvist, 2017). This cross-country comparison may point to the importance of considering both content and audience.

Second, the study offers insights into the role of foreign-sponsored mass media in post-conflict settings and suggests that informational outreach to populations marginalized under the previous regime can have important implications for nation rebuilding. The radio programs examined here challenged gender disparities, but similar platforms have also been used to promote a wide range of values and information. For example, the United States has operated Voice of America, its largest and longest-running international broadcaster, since 1942, and the BBC has long served as a global source of cross-border news. These precedents underscore the potential of foreign-sponsored media, as in my case, to influence social norms during critical periods of political and social reconstruction.

At the same time, my findings should not be interpreted as a blanket endorsement of foreign media intervention, nor as an argument for gender empowerment through military occupation. Rather, this study aims to understand how media exposure, under specific historical circumstances, affected women's outcomes.

[30]Examples include Woman's Hour on BBC Radio 4 in the UK (from 1946), Radio Donna in Italy (from 1976), Womankind in the US (1969), Hemmafru byter yrke in Sweden (1965), and RadiOrakel in Norway (from 1982).

Japan's case underscores the importance of local agency in shaping outcomes. As discussed in Section 2, women's programs were produced through collaboration between American and Japanese women. Although final approval rested with the occupying authorities, historians note that Japanese women played a central role in shaping and localizing the content, at times resisting imposed narratives. American contributors also made notable efforts to understand Japanese society and culture. The result was far from a unilateral imposition of norms. These production dynamics are crucial for assessing the functioning of foreign media policies, and future research should investigate the conditions under which such collaborations produce meaningful, locally resonant change.

8 Conclusion

This study examines the impact of women's radio programs broadcast by the US-led occupying force in Japan between 1945 and 1952 to dismantle prewar patriarchal norms. Using local variation in radio signal strength driven by soil conditions as an instrumental variable, I provide causal evidence that greater exposure to these programs increased women's electoral turnout. Radio exposure also accelerated fertility decline, not through changes in marriage rates, or greater use of induced abortions, but through the adoption of family planning. This decline benefited women by reducing health risks and child-rearing burdens and carried broader social implications. By contrast, I find no effect on female employment, which may reflect the tightened labor market following the return of men from World War II or the persistence of traditional attitudes among male-dominated employers. The latter possibility suggests that, because the radio campaigns were directed primarily at women, they may have overlooked the need to shift men's norms as well.

Taken together, my findings suggest that pro-gender-equality messages directed at women have empowered Japanese women in the aftermath of World War II. This not only has significant implications for women's lives, but also impacts society by shifting the political landscape and accelerating fertility transition, potentially paving the way for the rapid economic growth that followed. Thus the empowerment of women concerns not just women but society at large.

My findings open new avenues of economic research. First, the case of Occupied

Japan limits my ability to investigate what would have happened if both men and women, or only men, had been exposed to women's radio content. This limitation, however, provides motivation for field experiments to understand the nature of targeted mass media interventions further. Second, women who are exposed to new gender norms may have passed them onto their children even if they themselves did not change their behavior. Such an inter-generational transmission of the impact of norm-based interventions would be an interesting avenue of future research.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT 4o in order to check grammar and spelling. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix for

Empowering Women Through Radio: Evidence from Occupied Japan

Yoko Okuyama

Appendix Contents

A Data

A.1 Data Sources

Table A.1: Data Sources

Data source	**Variable**
Japan Broadcasting Corporation Statistic Report 1946	Post-war radio subscription rate No. of households in 1946
A map on medium wave field strength published in 1949	Field strength
Latitudes and longitudes of transmitters obtained from Japan Broadcasting Corporation Yearbook 1947 District boundaries year by year provided by Maruyama Lab, Tsukuba University	Distance to a nearby transmitter
Prefecture Annual Statistics Book (annually from 1949 to 1960)	Postwar crude birth rate
The Annual Vital Statistics Report in 1935	Marriages, divorces, stillbirths, and live births
The Annual Vital Statistics Report in 1947	Live births
The Annual Vital Statistics Report in 1950	Marriages, divorces, stillbirths, and live births
Population Census 1940	Prewar female employment rate, Sector labor share, male-to-female ratio
Population Census 1947	Population in 1947
Population Census 1950	Labor force participation Industrial composition
The 16th House of Representatives election results	Turnout in the 1928 election (men), the vote share of WSL-endorsed candidates
The 20th House of Representatives election results	Turnout in the 1928 election (men), the vote share of IRRA-endorsed candidates
The 21st House of Representatives election results	Turnout in the 1942 election (men)
News papers	Turnout in the 1946 election by sex
The 22nd House of Representatives election results A list of female candidates provided by Ito (2008)	Female candidate's vote share and characteristics
The 24th House of Representatives election results	Turnout in the 1949 election by sex
The 25th House of Representatives election results	Turnout in the 1952 election by sex
Overall Report of Damage Sustained by the Nation During the Pacific War	District-level total casualty during the World War II
The 1st Survey on the National Character of the Japanese (1953)	Radio listenership, gender attitudes

A.2 Mean Observable Characteristics

Table A.2: Mean Observable Characteristics

	(1) All mean/[std.dev.]	(2) In-sample prefectures mean/[std.dev.]	(3) Out-of-sample prefectures mean/[std.dev.]	(4) Diff. diff/(std.err.)
Radio subscription rate	0.373	0.381	0.364	0.017
	[0.124]	[0.122]	[0.127]	(0.01)
Field Strength (mv/m)	2.948	3.146	2.740	0.407
	[3.132]	[3.239]	[3.007]	(0.24)
Residualized field strength	0.000	-0.005	0.006	-0.011
	[0.410]	[0.404]	[0.417]	(0.03)
Distance from a nearby transmitter (km)	32.603	33.426	31.740	1.686
	[21.197]	[22.546]	[19.681]	(1.64)
No. of households (in 1,000)	1.874	1.891	1.856	0.035
	[1.695]	[1.878]	[1.480]	(0.13)
Large-scale civilian casualties in WWII	0.210	0.227	0.192	0.035
	[0.408]	[0.420]	[0.394]	(0.03)
Observations	662	339	323	662

Note: Summary statistics for the main independent variable (district-level radio subscription rate), IV (residualized field strength), and control variables. Column 1 include 784 districts but not 785 districts because one district (Hidaka *shicho*) is missing in the 1950 population census. Residualized field strength is computed by regressing field strength on the distance to a nearby transmitter and transmitter fixed effects. Male to female ratio is defined as the number of men per woman. In-sample prefectures (Column 2) are the ones for which I found turnout data. The complete list of the in-sample prefectures are as folllows: Iwate, Miyagi, Yamagata, Fukushima, Ibaraki, Tochigi, Gunma, Saitama, Chiba, Tokyo, Niigata, Ishikawa, Yamanashi, Nagano, Shizuoka, Aichi, Mie, Kyoto, Shimane, Yamaguchi, Tokushima, Ehime, Saga, Nagasaki, Oita and Miyazaki. The remaining 20 prefectures are the out-of-sample prefectures (Column 3). Column 4 shows the differences in means between the in-sample and out-of-sample prefectures. Standard deviations are in square brackets. Standard errors are in parentheses.

Source: Radio subscription rate, and the number of households come from Japan Broadcasting Corporation Statistic Report 1946; industrial compositions and the male to female ratio come from the Japan Population Census 1950; I construct an indicator variable for large-scale casualties based on a list of heavily-damaged cities in "Overall Report of Damage Sustained by the Nation During the Pacific War". Note that "Overall Report of Damage Sustained by the Nation During the Pacific War" reports May-1948 estimates, and is known to underestimate casualties. In my analyses, I assume that a potential measurement error is orthogonal to my instrumental variable.

B Balance Test

B.1 Variables and Definitions

Interwar Political Variables The interwar political variables, measured during the expansion of male suffrage and the rise of the women's suffrage movement, include electoral turnout in the 1928 general election (the first after full male suffrage), the presence of male candidates seeking endorsement from the Women's Suffrage League (WSL), the vote share of WSL-endorsed candidates conditional on having at least one, and turnout in the 1937 general election (the last before World War II). **Turnout in 1928 and 1937** serves as a measure of a district's political engagement and the penetration of democratic practices during the interwar period.

Data on the Women's Suffrage League provide a way to measure district-level support for women's political participation. In prewar Japan, and especially during the interwar period, the women's suffrage movement gained momentum under the leadership of the WSL, the largest non-partisan women's suffrage organization. One of the WSL's main activities in the 1928 general election, which was the first held after universal male suffrage, was to increase the number of pro-suffrage male representatives in the Lower House. To this end, it officially endorsed 14 male candidates across the political spectrum who supported women's suffrage: Fujimori Narukichi (Labour-Farmer Party); Hata Momosaku and Kamekichi Takahashi (Japan Farmers Party); Kato Choichi and Kosaka Junzo (Constitutional Democratic Party); Katayama Tetsu, Abe Isoo, Suzuki Bunji, Miyazaki Ryusuke, and Tamefuji Goro (Social Democratic Party); Hoshijima Jiro and Mikami Hidel (Rikken Seiyukai, Association of Friends of Constitutional Government); Sugiyama Genjiro (Japan Labour-Farmer Party); and Kaino Eizaburo (Jitsugyo Doshikai, Business Men's Party). At the time, public support for women's suffrage was stigmatized as a feminine and thus politically undesirable position for male politicians. The fact that these candidates actively sought WSL endorsement, and that seven of them were elected, suggests that their constituencies were more supportive of women's suffrage than others. Based on this, I construct two measures of district-level support for women's suffrage: an indicator equal to one if at least one WSL-endorsed candidate ran in the district, and, conditional on having such a candidate, the share of votes received by WSL-endorsed candidates.

Wartime Political Variables The wartime political variables, reflecting the period of rising militarism, include electoral **turnout** and **the vote share of the Imperial Rule Assistance Association** (IRAA; *Taisei Yokusan Kai*) **in the 1942 wartime election**, drawn from official election data. Established in October 1940 by the prime minister, the IRAA sought to replace existing political parties, unify the population under a single organization, and mobilize support for Japan's war in Asia. Although it absorbed recognized parties, it never achieved the centralized control of its Nazi German counterpart. Business and financial sectors resisted its authority, and critics argued it undermined imperial constitutional principles. In the 1942 election, non-endorsed candidates still won seats in the Imperial Diet. Over time, the IRAA became primarily an instrument of Home Ministry control, despite attempts by the army to direct its activities. I measure district-level support for the IRAA, and thus for the military government, by the share of votes received by IRAA-endorsed candidates. Every district had at least one endorsed candidate, so all districts are included in the sample.

Economic and Demographic Variables Economic and demographic variables are measured around the onset of the war, and include **the male-to-female ratio** at birth, draw from the vital statistics in 1935, the overall **male-to-female ratio** drawn from the 1940 population census, and the extent of **gender-based industrial segregation measure**, Theil H index, an entropy-based dissimilarity measure from information theory, using data from the 1940 census, which reports the number of men and women employed in nine broad industries: Agriculture, Fishery, Mining, Manufacturing, Commerce, Transportation, Public Sector, Housekeeping, and Others. No finer industry breakdown is available at the district level. **The labor share in the primary sector** is computed based on the 1940 population census.

Pre US Occupation Outcomes of Interests The **marriage rate**, defined as the number of marriages per 1,000 population, and the **birth rate**, defined as the number of births per 1,000 population, are taken from the 1935 vital statistics. The **female employment rate**, drawn from the 1940 population census, excludes family employment.Women's turnout prior to the occupation period is unavailable, as women did not yet have voting rights.

To assess whether the instrument provides variation that is as good as random, I examine whether the residualized field strength is uncorrelated with key pre-war (pre-women's radio) variables. Specifically, I first regress the radio signal on the baseline set of covariates and compute the residuals, which capture the identifying variation in the instrument. I then test whether this residualized radio signal correlates with pre-war variables that could also be related to my outcome of interest. All variables show no correlation with the residualized field strength, supporting the validity of the instrument.

Table B.3: Additional Balance Test

	(1) Sub Sample	(2) All Sample	(3) All Sample
Turnout in 1942	-0.0808	0.970	-0.239
	(0.580)	(1.342)	(0.799)
IRAA Vote share in 1942	0.0569	0.0805	0.105
	(0.115)	(0.189)	(0.123)
Turnout in 1928	0.798	1.135	0.656
	(0.636)	(1.015)	(0.490)
= 1 if any WSL endorsed candidate in 1928	-0.0606	-0.0886	-0.0458
	(0.0760)	(0.0693)	(0.0909)
Turnout in 1937			0.438
			(0.546)
Primary sector labor share in 1940			-0.242
			(0.265)
Gender industrial segregation in 1940			0.125
			(0.713)
Female employment rate in 1940			-0.0899
			(0.652)
No. live births per 1000 population in 1935			0.00180
			(0.00785)
No. marriages per 1000 population in 1935			0.0286
			(0.0231)
Male to female ratio in 1940			-0.00321
			(0.00370)
Male to female ratio at birth in 1935			-0.00605
			(0.00485)
R2	0.01	0.04	0.03
Observations	662	339	634

Note: WSL – Women's Suffrage League; IRAA – Imperial Rule Assistance Association. This table presents regression results examining the relationship between pre-war district characteristics and the residual radio signal. The residual signal is constructed by regressing radio field strength on decile-binned distance to the nearest transmitter, radio station fixed effects, prefecture fixed effects, the logarithm of the number of households, and an indicator for severe WWII damage from air raids or the atomic bomb. This residual captures variation in radio signal strength attributable to soil conditions between the transmitter and the receiving district. The unit of observation is a district (*shi* or *gun*), using 1950 boundaries. In Panel A, Column 3 restricts the sample to districts with at least one WSL-endorsed candidate. Column 4 restricts the sample to districts where voting occurred, excluding districts where the number of candidates equaled the number of seats. All regressions are weighted by district population, and standard errors are clustered at the radio station level.

C Regression Analyses

C.1 Robustness

Table C.4: Impacts of Women's Radio: Robustness

	Women's Turnout 1946	Men's Turnout 1946	Female Employment Rate 1950	Marriage Rate 1950	Divorce Rate 1950	Birth Rate 1947	
	(1) TSLS	(2) TSLS	(3) TSLS	(4) TSLS	(5) TSLS	(6) TSLS	(7) TSLS
Radio subscription in std.dev. unit	0.0242**	0.00835	0.00132	0.0308	0.00792	-0.300	-1.550**
	(0.00763)	(0.00695)	(0.0115)	(0.178)	(0.0364)	(0.379)	(0.509)
Turnout 1942	0.238*	0.600***					
	(0.103)	(0.0686)					
Female Employment Rate 1940			0.403***				
			(0.0900)				
No. of marriages per 1000 population				0.0288			
				(0.0586)			
No. of divorces per 1000 population					0.409***		
					(0.114)		
Birth Rate 1935						0.325***	0.396***
						(0.0402)	(0.0502)
Observations	339	329	658	660	659	662	662
First stage F-stat	29.23	28.22	31.46	34.22	33.58	31.99	31.99
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.67	0.81	0.28	8.42	0.97	34.54	28.14

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: TSLS = two-stage least squares

Table C.5: Impacts of Women's Radio: Robustness

	Women's Turnout 1946	Men's Turnout 1946	Female Employment Rate 1950	Marriage Rate 1950	Divorce Rate 1950	Birth Rate 1947	
	(1) TSLS	(2) TSLS	(3) TSLS	(4) TSLS	(5) TSLS	(6) TSLS	(7) TSLS
Radio subscription in std.dev. unit	0.0209**	0.0114	-0.0159	0.304	-0.0400	-0.287	-1.545**
	(0.00749)	(0.00758)	(0.00880)	(0.232)	(0.0398)	(0.404)	(0.520)
Turnout 1942	0.251*	0.596***					
	(0.105)	(0.0855)					
Female Employment Rate 1940			1.006***				
			(0.0879)				
No. of marriages per 1000 population				-0.0310			
				(0.0691)			
No. of divorces per 1000 population					0.431***		
					(0.111)		
Birth Rate in 1935						0.325***	0.382***
						(0.0391)	(0.0512)
Male to Female Ratio in 1940	0.000479	-0.000213	-0.00393***	0.0403**	0.00119	0.0627**	0.0823**
	(0.000512)	(0.000577)	(0.000656)	(0.0143)	(0.00257)	(0.0229)	(0.0258)
Male to Female Ratio at Birth in 1935	0.00000846	-0.000159	0.0000900	0.00869	-0.000517	0.00776	0.0143
	(0.000550)	(0.000446)	(0.000291)	(0.00841)	(0.00178)	(0.0150)	(0.0180)
Industrial Sex Segregation Index in 1940	-0.00723	0.0243	0.0996	-2.427	-0.165	3.861	-0.827
	(0.103)	(0.118)	(0.0564)	(1.558)	(0.733)	(3.276)	(3.886)
Primary Sector Labor Share in 1940	-0.0583**	0.0387*	-0.148***	1.403*	-0.385***	-0.742	-1.043
	(0.0218)	(0.0181)	(0.0202)	(0.694)	(0.105)	(0.940)	(0.882)
Observations	336	326	658	656	655	658	658
First stage F-stat	25.76	24.20	27.81	26.26	26.39	26.37	26.37
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.67	0.81	0.28	8.42	0.97	34.54	28.14

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: TSLS = two-stage least squares

C.2 Spatial Standard Errors

Table C.6: Impacts of Women's Radio: Conley Standard Errors

	Women's Turnout 1946	Men's Turnout 1946	Female Employment Rate 1950	Marriage Rate 1950	Divorce Rate 1950	Birth Rate 1947	
	(1) TSLS	(2) TSLS	(3) TSLS	(4) TSLS	(5) TSLS	(6) TSLS	(7) TSLS
Radio subscription in std.dev. unit	0.0242**	0.00967	-0.00559	0.0282	0.0127	-0.710	-2.064***
	(0.00771)	(0.00729)	(0.00842)	(0.155)	(0.0308)	(0.410)	(0.457)
Observations	335	325	658	656	655	658	658
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.67	0.81	0.28	8.42	0.97	34.54	28.13

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: TSLS = two-stage least squares

C.3 Heterogeneous Effects

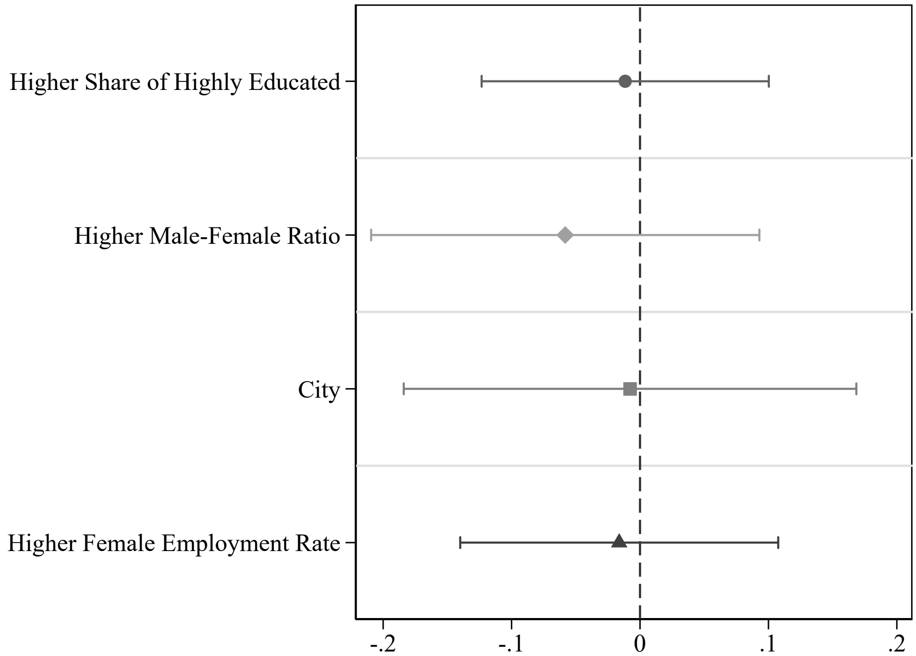


Figure C.1: Differences in Mean Residual Radio Signal Across Socioeconomic and Demographic Indicators

Notes: This figure presents differences in the mean residual radio signal across socioeconomic and demographic indicators: whether a district had an above-median share of highly educated individuals in 1950, an above-median male-to-female ratio in 1950, was classified as a city, or had an above-median prewar female employment share. Each difference is estimated as the coefficient from a separate regression of the residual radio signal on the respective indicator. Regressions are weighted by district population, and standard errors are clustered at the radio transmitter level. Whiskers indicate 95% confidence intervals.

Table C.7: Impacts of Women's Radio: Heterogeneous Effects on Women's Turnout

	Women's Turnout				
	(1) TSLS	(2) TSLS	(3) TSLS	(4) TSLS	(5) TSLS
Radio subscription in std.dev. unit	0.0246**	0.0234**	0.0141*	0.0304***	0.0220*
	(0.00807)	(0.00734)	(0.00701)	(0.00879)	(0.0101)
More highly educated women (above median)		0.000440			
		(0.0321)			
More highly educated women × Radio subscription (std.dev.)		0.00134			
		(0.00968)			
City = 1			-0.0335		
			(0.0449)		
City × Radio subscription (std.dev.)			0.0191		
			(0.0121)		
Higher male to female ratio (above median)				0.0593*	
				(0.0269)	
Higher male to female ratio × Radio subscription (std.dev.)				-0.0269***	
				(0.00764)	
Higher Female Employment (above median)					-0.0392
					(0.0314)
Higher Female Employment × Radio subscription (std.dev.)					0.00534
					(0.00948)
First stage F-stat	28.26	16.28	14.16	16.44	14.40
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.67	0.67	0.67	0.67	0.67
Observations	339	339	339	339	336

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: TSLS = Two-Stage Least Squares. This table reports TSLS estimates of the effect of radio exposure on women's turnout, along with heterogeneous effects by district characteristics. Radio exposure is instrumented using field strength. All regressions control for nearest transmitter fixed effects, prefecture fixed effects, deciles of distance to the nearest transmitter, and log population size. Standard errors are clustered at the nearest radio transmitter level.

Table C.8: Impacts of Women's Radio: Heterogeneous Effects on Birth Rate

	Birth Rate				
	(1) TSLS	(2) TSLS	(3) TSLS	(4) TSLS	(5) TSLS
Radio subscription in std.dev. unit	-2.034***	-1.566**	-1.628**	-1.506**	-2.338***
	(0.514)	(0.576)	(0.501)	(0.562)	(0.646)
More highly educated women (above median)		1.398			
		(1.368)			
More highly educated women × Radio subscription (std.dev.)		-0.869*			
		(0.392)			
City = 1			-1.291		
			(1.571)		
City × Radio subscription (std.dev.)			-0.326		
			(0.386)		
Higher male to female ratio (above median)				2.251	
				(1.523)	
Higher male to female ratio × Radio subscription (std.dev.)				-0.293	
				(0.530)	
Higher Female Employment (above median)					-1.728
					(1.239)
Higher Female Employment × Radio subscription (std.dev.)					0.679
					(0.369)
First stage F-stat	36.41	11.41	18.22	22.73	14.72
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes
Distance control	decile bins	decile bins	decile bins	decile bins	decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes
Mean outcome	28.14	28.14	28.14	28.14	28.14
Observations	662	662	662	662	658

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: TSLS = Two-Stage Least Squares.

D Analysis of the 1928 Election Turnout

In Section 6, I show that radio exposure increased women's turnout but had no significant effect on men's turnout in the 1946 House of Representatives election. As I argue in Section 7.2, this pattern suggests that the effect on women was driven primarily by content targeted at them (i.e., women's programs) rather than by exposure to radio broadcasts in general (e.g., news programming). However, this interpretation relies on the assumption that men and women were equally susceptible, or equally unaffected, by general information transmitted through radio. This assumption may not hold if newly enfranchised voters are particularly more responsive to radio than voters who have had prior voting experience. This concern might be especially relevant in the 1946 context, where all women were voting for the first time, while men over the age of 25 had been eligible to vote since the prewar period.

To examine whether radio exposure itself contributes to mobilizing *newly enfranchised* voters, I turn to the 1928 House of Representatives election. This case is useful for three reasons. First, it was the first general election after the 1925 reform that extended voting rights to all adult men, removing the previous income-based eligibility requirement. Second, radio broadcasting had started in Japan in 1925, so the 1928 election was also the first nationwide election held after voters gained access to this new mass medium. Thus, the majority of voters in 1928 were newly enfranchised and voting under novel media conditions. Third, unlike in 1946, there was no targeted messaging aimed specifically at newly enfranchised voters in 1928. Therefore, the 1928 election provides a compelling setting to test whether general radio access, absent targeted content, can influence turnout among newly enfranchised voters.

I estimate the reduced-form impact of residualized field strength on voter turnout in the 1928 election. To construct the residualized field strength measure, I follow the same approach as in the balance tests: I regress observed field strength on controls such as distance to the nearest transmitter and transmitter fixed effects, and extract the residual variation attributable to soil conditions. I then regress district-level turnout in 1928 on this residualized measure of field strength.[31] Conceptually, this tests whether men's turnout in 1928 was higher in areas with better natural conditions

[31] This approach rests on the assumption that the soil-related factors affecting radio signal reception remained stable between 1928 and the U.S. occupation period.

Table D.9: Impacts of Radio: Prewar Turnout

	Men's turnout 1928	Men's turnout 1937
	(1)	(2)
	RF	RF
Residualized Field Strength	0.00857	-0.0000166
	(0.00559)	(0.00687)
Observations	662	638
Mean outcome	0.82	0.78

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: RF = Reduced Form.

for radio reception.[32]

Table D.9, Column 1 shows that the estimated coefficient is close to zero and not statistically significant. To examine potential lagged effects of radio exposure, I replicate the analysis using turnout in the 1937 election, nearly a decade later. Again, the estimated effect is nearly zero and not statistically significant. These findings suggest that general access to radio did not significantly increase turnout in the settings I examine, even when a large share of the electorate had recently gained the right to vote, and radio was new mass medium.

[32] Due to the absence of district-level data on radio listenership in 1928, I cannot provide a two-stage least squares (TSLS) estimate.

E Historical Background

E.1 Radio Subscription Fee

Table E.10: Radio subscription fees

	Radio subscription fee (monthly)	TV subscription fee (monthly)	Starting teacher salaries (monthly)	Japanese soba noodle unit price
1925	1.00		45.00	0.10
1930	0.75		45.00	0.10
1933	0.75		55.00	0.10
1937	0.50		55.00	0.13
1941	0.50		55.00	0.16
1946	2.50		400.00	-
1948	35.00		2000.00	-
1950	35.00		5000.00	15.00
1954	67.00	300.00	7800.00	30.00
1955	67.00	300.00	7800.00	30.00

Notes: This table reports the monthly radio subscription fee (in Japanese Yen) from 1925 to 1955 along with the monthly salary for first-year teachers and a unit price for Japanese soba noodles. While inflation accelerated, the JBC decreased the monthly fee and made radio more accessible for a wide income-range of Japanese: the fee was 1 Japanese yen compared to 45 yen of teachers' starting salaries in 1925. In 1941, however, the fee halved while teacher salaries moderately grew. After World War II, inflation outpaced the increase in the nominal subscription fee, and thus the subscription fee further drops in real terms. Therefore, I am less concerned that subscription fees deterred low income Japanese from listening to the radio. Source: Okabe (2018) *"The 50 Years of Japanese Radio 1925-1975"* The Japan Radio Museum.

Table E.10 shows the monthly radio subscription fee (in Japanese Yen) from 1925 to 1955 along with the monthly salary for first-year teachers and a unit price for Japanese soba noodles. While inflation accelerated, the JBC decreased the monthly fee and made radio more accessible for a wide income-range of Japanese: the fee was 1 Japanese yen compared to 45 yen of teachers' starting salaries in 1925. In 1941, however, the fee halved while teacher salaries moderately grew. After World War II, inflation outpaced the increase in the nominal subscription fee, and thus the subscription fee further drops in real terms.

E.2 Analyzing radio content

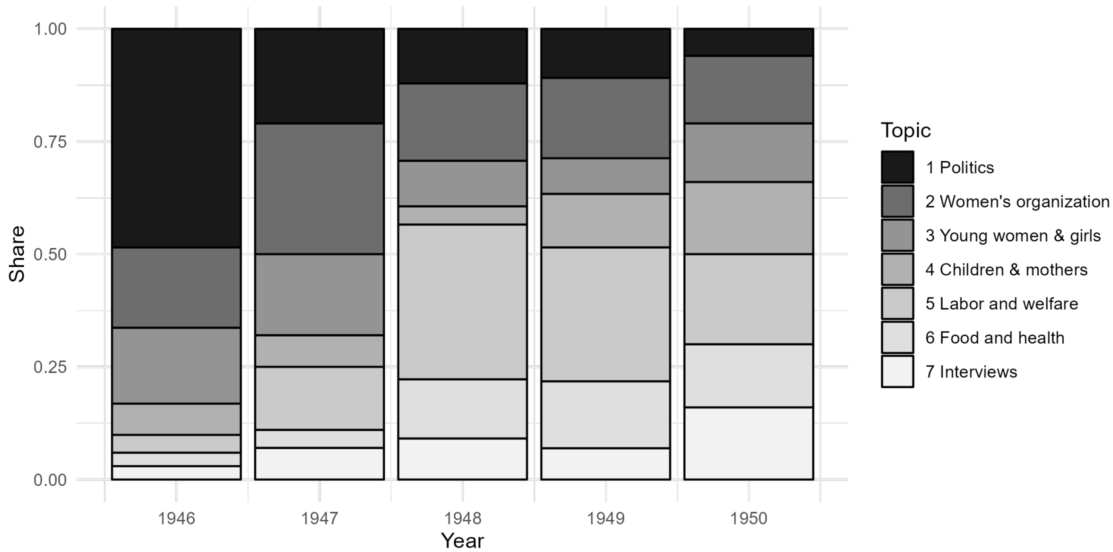


Figure E.2: Topic Composition of Women's Radio

Notes: This figure presents topic composition of women's radio programs from 1946 to 1950. Contents descriptions are drawn from GHQ/SCAP CIE Weekly Report (Radio Education Branch, 1946-1950, GHQ/SCAP CIE Box No. 5318 Folder 9) and classified using Latent Dirichlet allocation.

Latent Dirichlet Allocation (LDA) is an unsupervised machine learning technique for topic modeling. It considers each document as a predetermined number of topics in a certain proportion, and each topic as a collection of keywords in a certain proportion. A goal of LDA is to estimate a word distribution within each topic, then a topic distribution within each document by maximum likelihood. In other words, LDA tries to find a topic model that fits best to the corpus within a collection of documents under analysis. In my study, a document corresponds to a daily content of women's radio programs, two of which are quoted in Subsection 2.3. On the one hand, each topic obtains a probability distribution over words. Figure E.3 shows word clouds for each of seven topics; each word cloud shows the top-30 most frequently used words in each topic. The relative size of a word corresponds to its assigned probability, that is, larger words are weighted higher within the topic. Based on the word clouds, I assign labels to the topics: politics, women's organization, young women and girls, children and mothers, labor and welfare, food and health, and interviews. Each document (i.e., a daily content description) obtains a probability distribution over the seven topics as in Figure E.2. Then, for further simplicity, I assign each document one topic with the

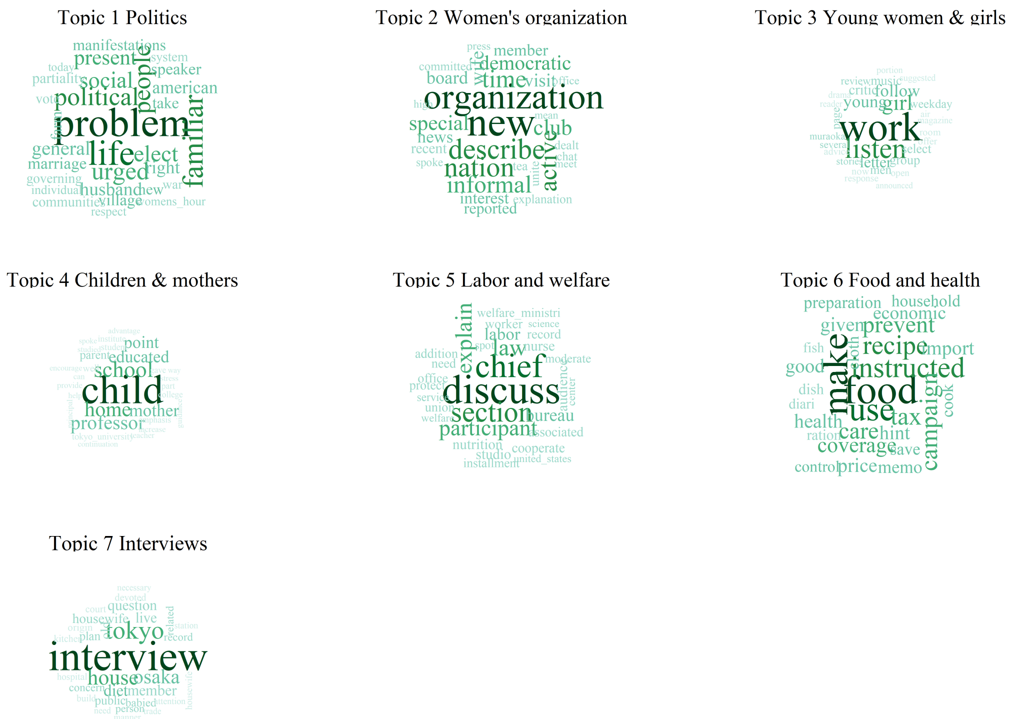


Figure E.3: Word Clouds

Notes: This figure presents word clouds characterizing each of seven topics in women's radio programs.

highest probability, and obtain a mapping from a collection of documents to the seven topics. The choice of seven groups was made as follows. On the one hand, in a model with a larger number of topics than seven, some topics appear to be too specific. On the other hand, in a model with a smaller number of topics than seven, some topics turn out to be too general and need to be split to be interpretable. However, my main takeaways are robust to the number of topics.

E.3 Airtime Allocation for Women's Programs

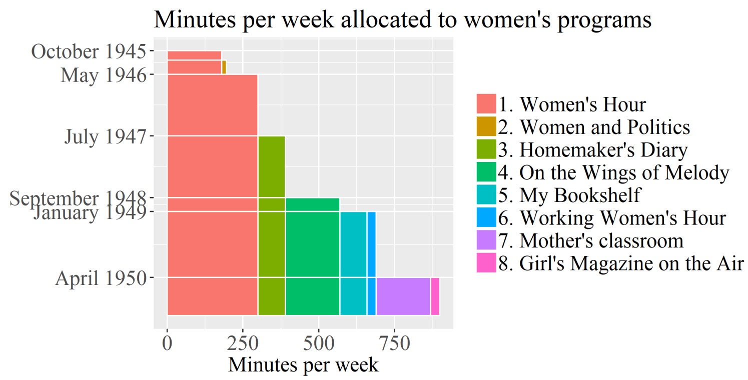


Figure E.4: Airtime Allocation

Notes: Thif figure presents airtime allocated to women's programs. The programs were categorized as "Women's program" by the JBC. Source: NHK Yearbook (1947, 1949) and GHQ/SCAP CIE Weekly Report (Radio Education Branch, 1946 - 1950).

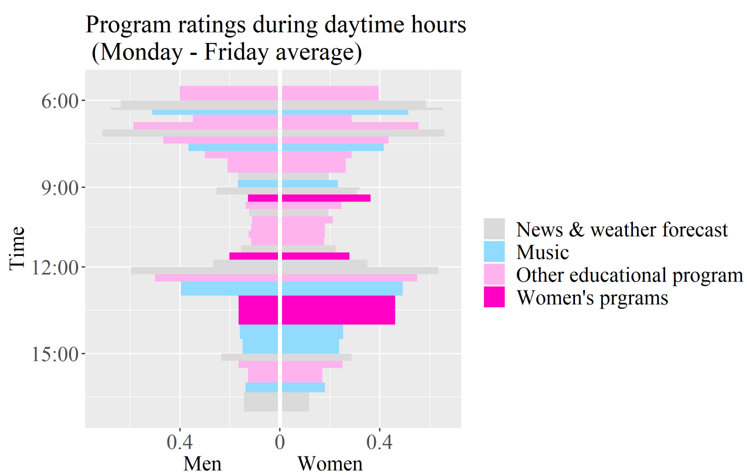


Figure E.5: Program Ratings During Daytime hours

Notes: This figure presents Channel 1 program ratings during daytime hours. Rating is defined as a percentage of male and female radio holders who actively listened to the radio in a given time slot. The figure shows the average rating from Monday to Friday in the survey week. Dark pink represents time slots allocated to women's programs, which were aired daily. Grey, pale blue, and pale pink indicate news and weather forecasts, music, and educational programs respectively. Men's rating is no larger than women's rating for any time slot. Women's rating is especially higher than men's for women's programs: the gender rating gap is 23.6 percentage points from 9:15 to 9:30, and 29.6 percentage points from 1 pm to 2 pm. Source: Radio Listeners Survey — Report of the 3rd Regular Survey, Part 1-4 (November 1948).

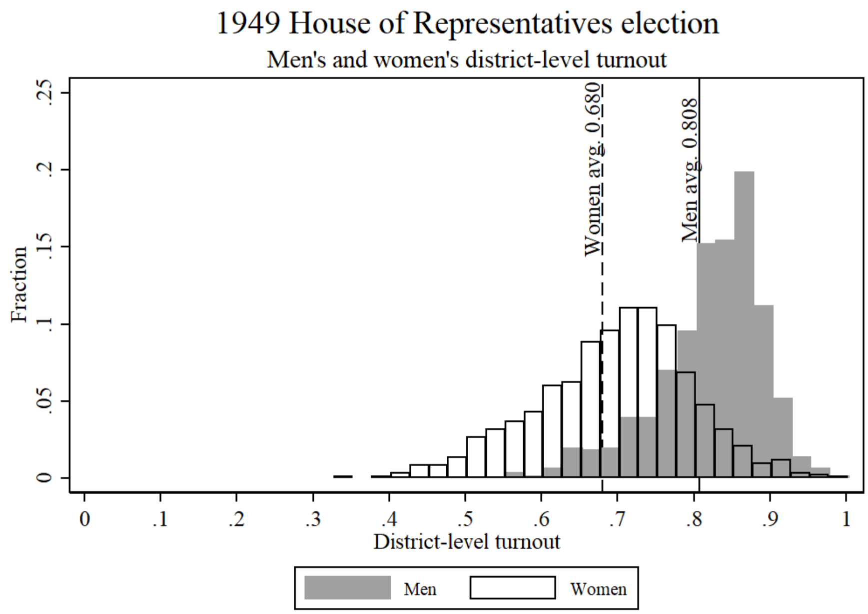
E.4 Electoral turnout in the 1949 and 1952 elections

In this section, I examine whether women's exposure to women's radio programs had persistent effects on turnout in later elections. I digitized data on women's and men's turnout in the 1949 (24th) and 1952 (25th) House of Representatives elections.[33] Figure E.6 shows histograms of turnout by gender. Women's turnout shifted upward between 1949 and 1952.

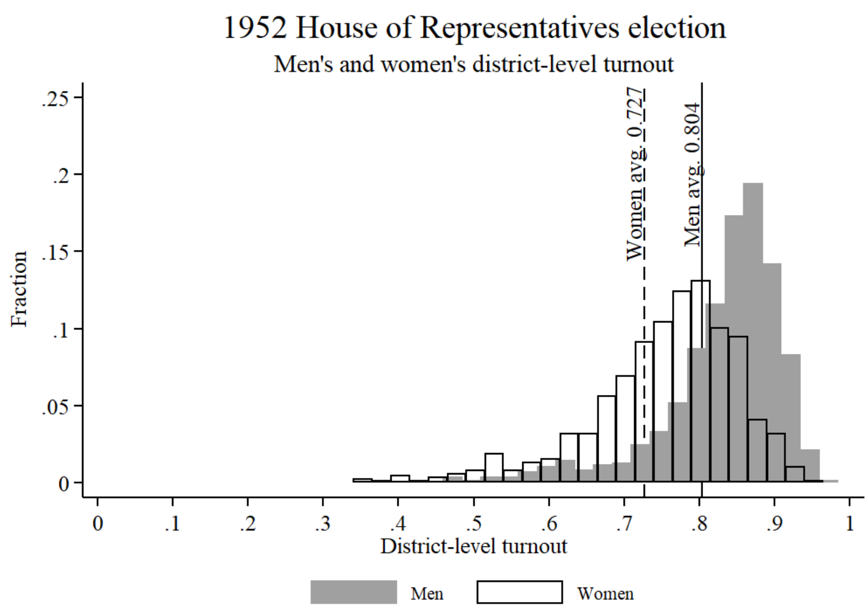
Estimating the long-run impact of radio exposure is complicated by evidence that the Stable Unit Treatment Value Assumption (SUTVA) may not hold after 1946. Contemporary newspapers and occupation authority records indicate intensified mobilization in low-exposure districts. For example, the Asahi newspaper publicly criticized Kojimachi district for low turnout after the 1946 election, a practice repeated in later elections. Archival records from the GHQ/SCAP (RG331, Box 2957) describe efforts by civil education officer Josephine Colletti in Hyogo Prefecture, including the establishment of the Tarumi League of Women Voters. Drawing on the model of the American League of Women Voters, such leagues likely organized house-to-house visits, get-out-the-vote campaigns, and citizenship schools.

Consistent with these accounts, Figure E.7 shows that low-exposure districts saw greater increases in women's turnout, narrowing the gap with high-exposure areas. This suggests spillover effects through inter-district competition. While these dynamics limit causal inference on long-run impacts, they illustrate how social change can diffuse spatially. By the 1960s, the gender turnout gap had closed, with men's and women's rates moving together thereafter (Figure E.8).

[33] The April 25, 1947 election is excluded because gender-disaggregated turnout data are unavailable.



(a) 1949



(b) 1952

Figure E.6: District-level electoral turnout in the 1949 and 1952 House of Representatives elections. Averages are weighted by the number of eligible voters.

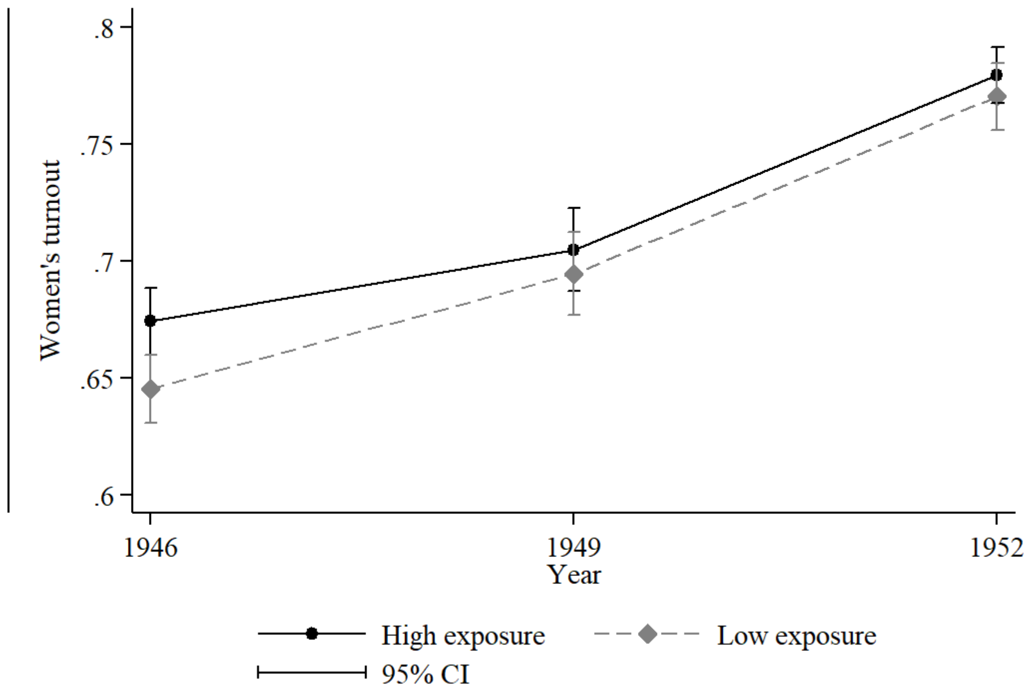


Figure E.7: The average women's turnout in the 1946, 1949 and 1952 elections by the degree of radio exposure. The radio exposure is measured by the 1946 radio subscription rate. Districts with above-the-median radio exposure are categorized as a high-exposure group, and vice versa. The vertical lines show confidence intervals. The 1947 election is not shown because the gender breakdown of district-level turnout is not available.

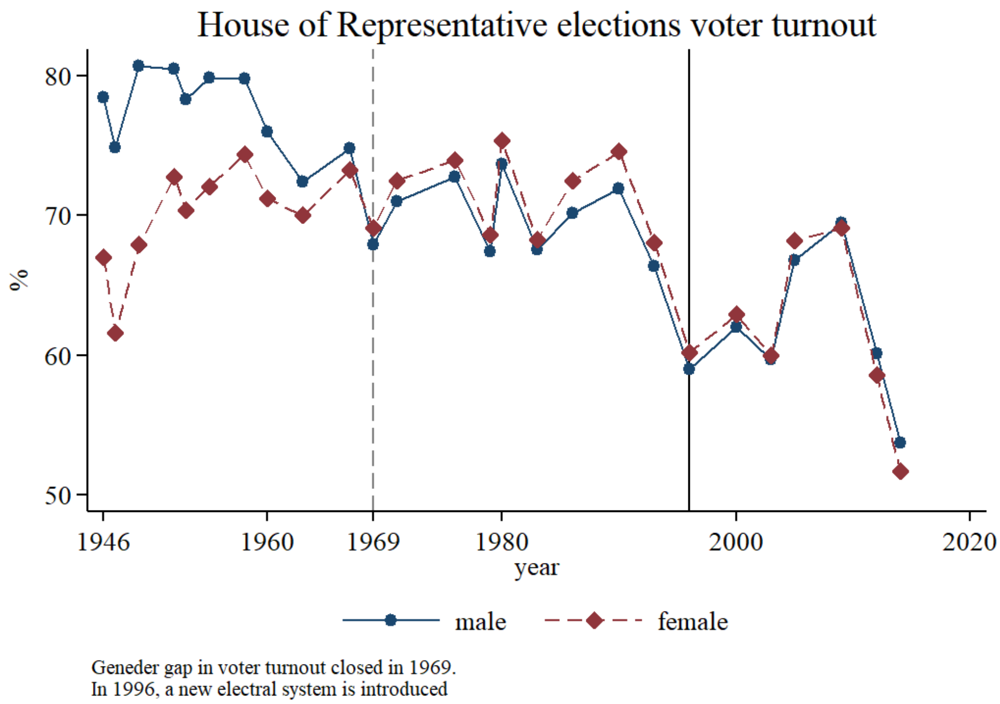


Figure E.8: House of Representatives Election Voter Turnout by Sex.

E.5 Historical demographic trends in Japan

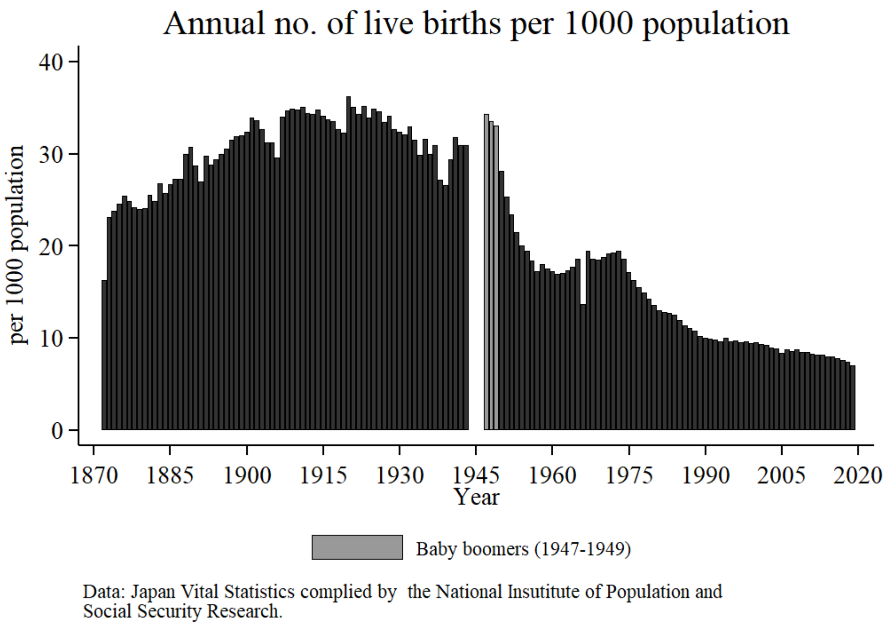


Figure E.9: No. of live births per 1000 population

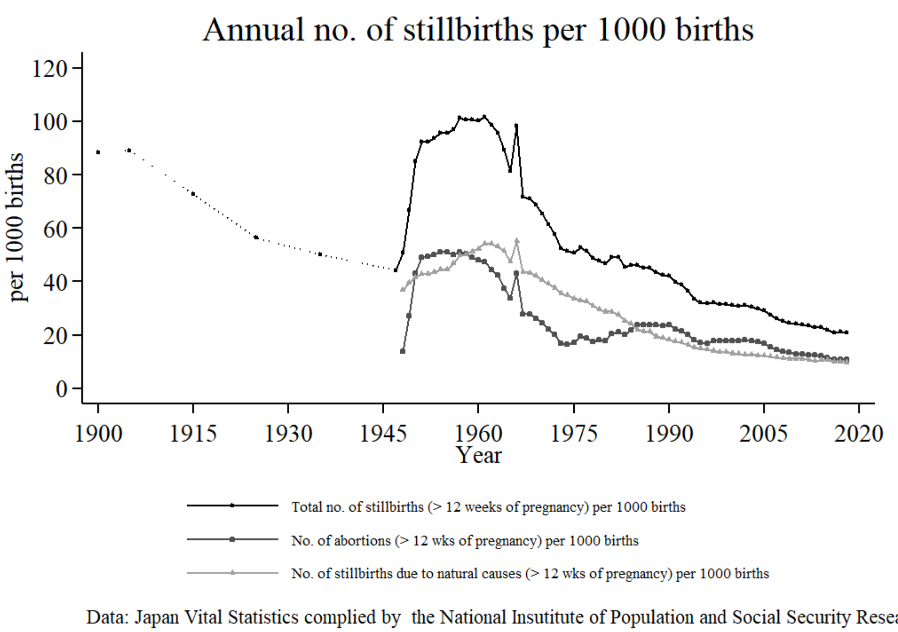


Figure E.10: No. of stillbirths per 1000 births

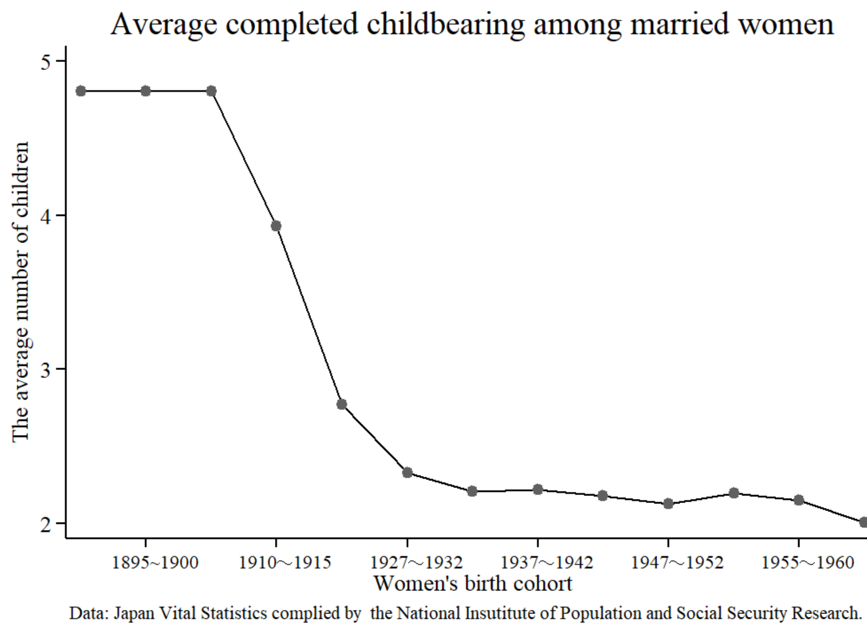


Figure E.11: Complete Birth Rate per Married Woman

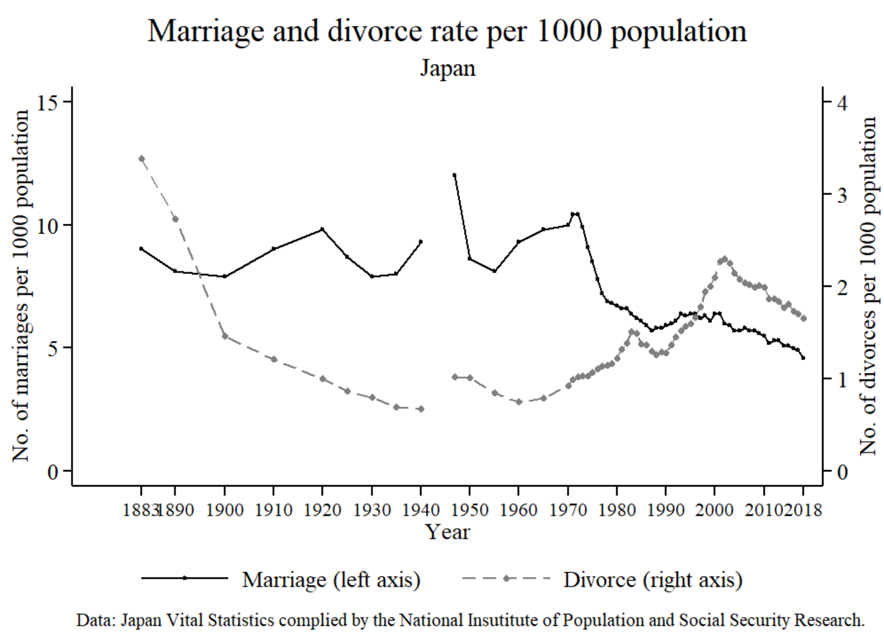


Figure E.12: Marriage and Divorce Rates

E.6 Radio Exposure and Gender Norm (1953)

Table E.11: Radio Exposure and Gender Norm from the 1953 Survey

	Approval of Women Working Outside	
	(1)	(2)
Radio:Female	0.090**	0.084*
	(0.045)	(0.044)
Radio	0.010	−0.015
	(0.034)	(0.034)
Female	−0.033	−0.026
	(0.035)	(0.035)
Left elementary school before graduation		−0.118
		(0.099)
Completed elementary school		0.002
		(0.080)
Higher elementary school		0.028
		(0.080)
New system: Junior high school		0.156
		(0.128)
Old system: Middle school		0.107
		(0.081)
New system: High school		0.218**
		(0.100)
Old system: Higher technical school		0.142
		(0.093)
New system: University		0.229
		(0.155)
Old system: University		−0.081
		(0.116)
Other (specify school/years)		−0.061
		(0.110)
No response		0.303
		(0.322)
(Intercept)	0.715***	0.683***
	(0.029)	(0.082)
Outcome mean	0.73	0.73
Observations	1,775	1,775
R^2	0.007	0.028

Notes: This table reports results from regressing a dummy variable that equals one if a respondent approves of women working outside the home on sex (*Female*), frequency of radio listening (*Radio*, coded as frequent listening versus only sometimes or not at all), and their interaction (*Radio:Female*). Column 2 additionally controls for education-level dummies. Within the education categories, *Old system* indicates schooling completed under the prewar education system, while *New system* refers to schooling completed under the postwar system. The reference group is respondents with no schooling. The data for this secondary analysis, "The 1st Survey on the National Character of the Japanese, 1953 [Database for Survey on Political Awareness, R-123], (Ichiro Miyake)" was provided by the Social Science Japan Data Archive, Center for Social Research and Data Archives, Institute of Social Science, The University of Tokyo. https://doi.org/10.34500/SSJDA.M165